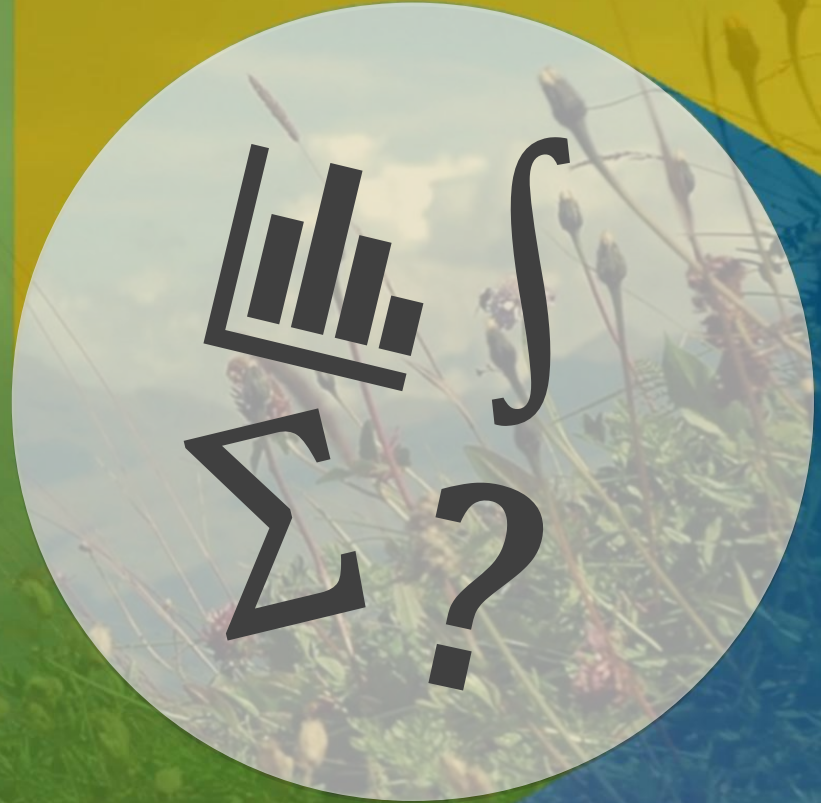


Introduction to Bayesian Statistics

Part 5 Generalized Linear Models

Benjamin Rosenbaum

iDiv 2026



This lecture

Introduction to GLM

Discrete responses

- Logistic regression
- Binomial regression
- Poisson regression

Continuous responses

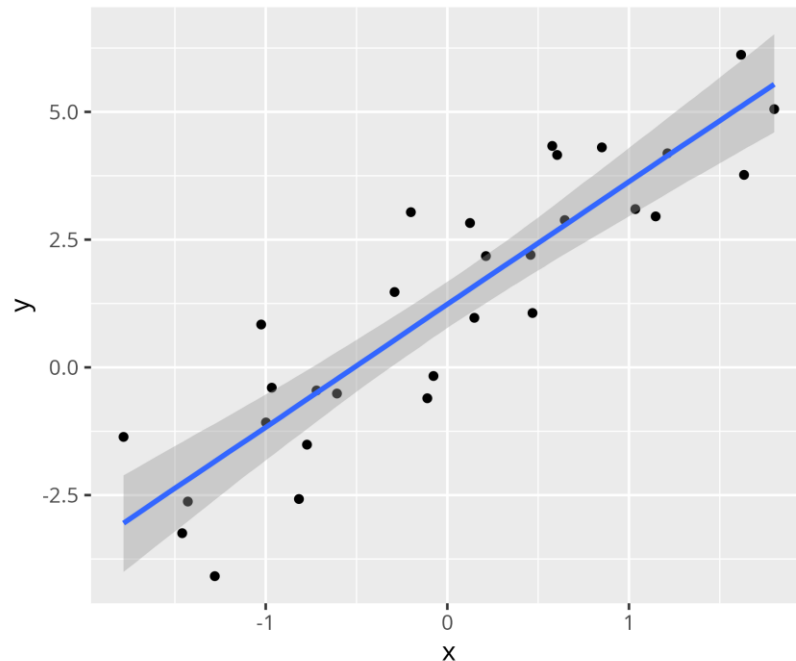
- Beta regression
- Distributional model

Nonlinear models

Summary

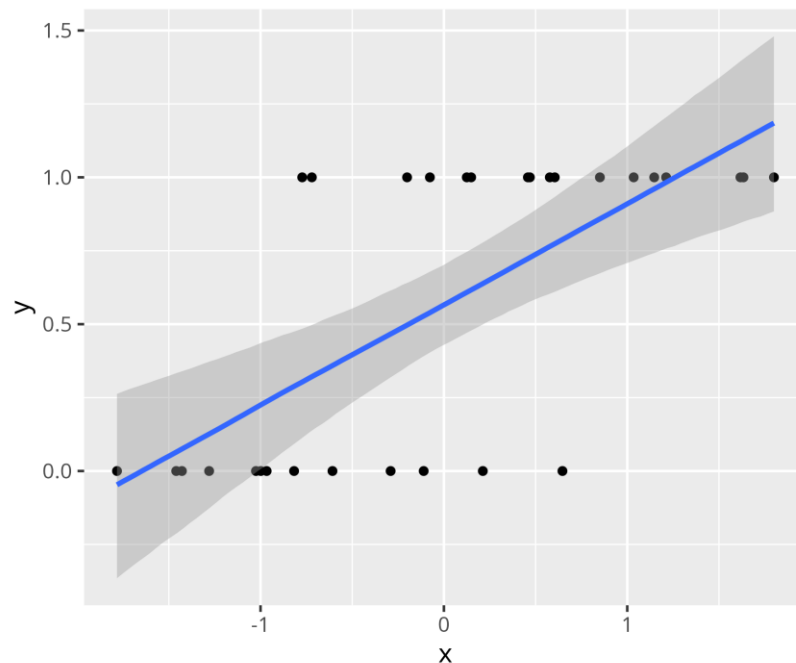
Assumptions for linear models

- 1. Linearity:** Trend of y follows linear prediction model $\mu(x) = a + b \cdot x$
- 2. Independence** of residuals:
not correlated (e.g. space or time)
- 3. Equal variance** of residuals:
across whole range of x
- 4. Normality** of residuals:
 $\varepsilon \sim \text{Normal}(0, \sigma)$



Assumptions for linear models

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What do we need?

Deterministic part:

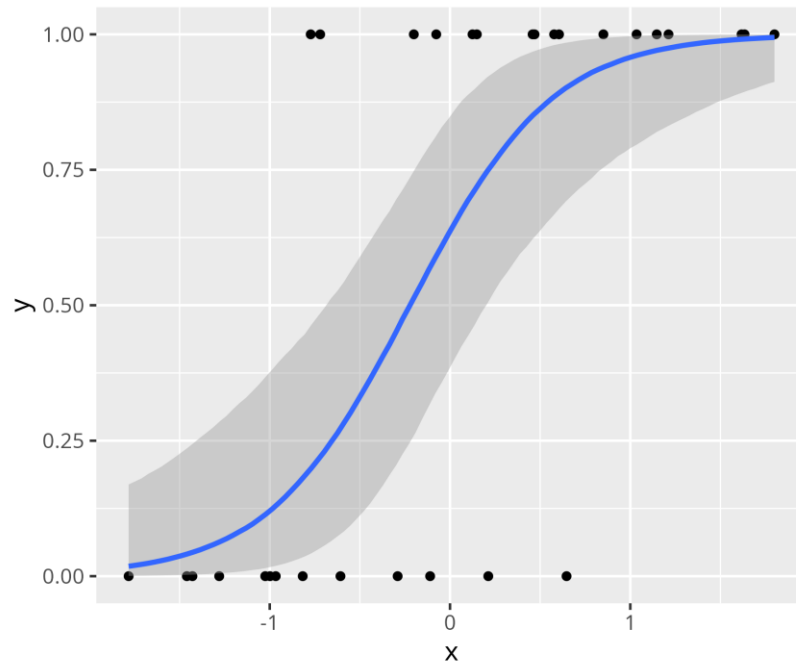
~~$\mu(x) = a + b \cdot x$~~

→ Regression curve $\mu(x)$ that respects actual boundaries of the data

Stochastic part:

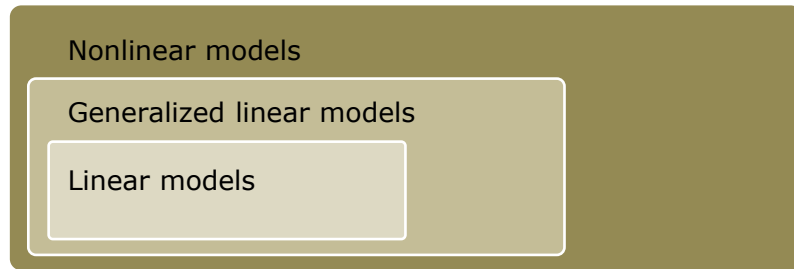
~~$y \sim \text{Normal}(\mu, \sigma)$~~

→ A distribution that models non-normal residuals correctly (discrete, skewed, non-constant var, ...)

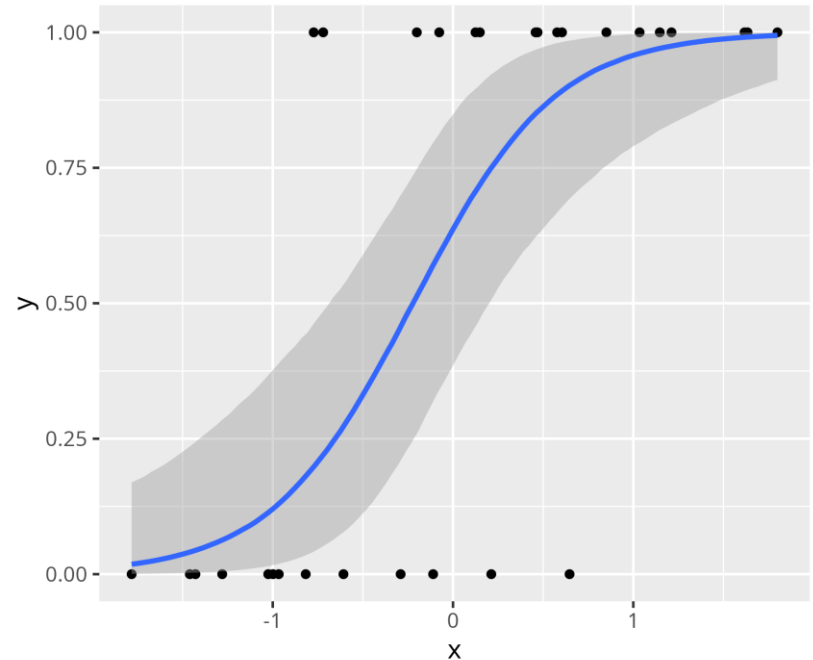


The generalized linear model (GLM)

Theoretically, we could fit all kinds of nonlinear regression curves $\mu(x)$ to the data



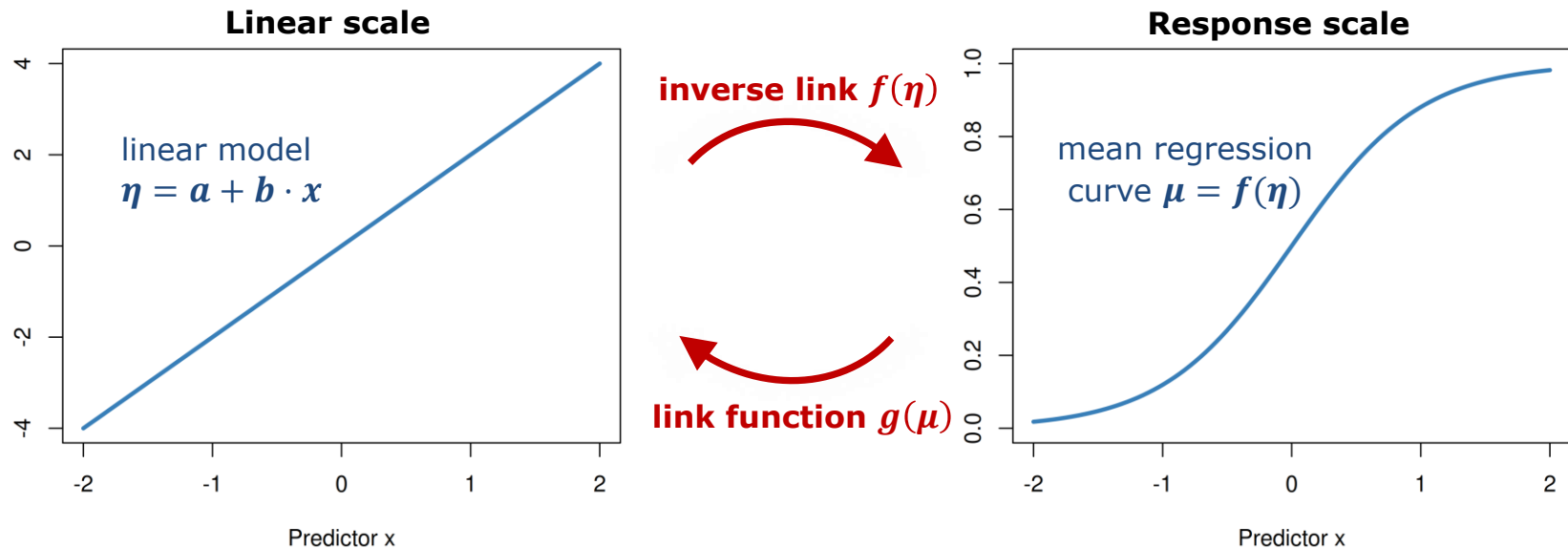
GLM is a special case of nonlinear models, that maps a linear model $\eta(x) = a + bx$ with a nonlinear function $\mu = f(\eta(x))$ to the response scale y



Why? „Easy“ to use.

Model structures, interactions, effect sizes, etc in the linear part are meaningful

Deterministic part: the link function g



Inverse link *compresses* linear model $\eta(x)$ to response scale

Link function *expands* mean regression curve $\mu(x)$ to linear scale

(Yes, this notation is very confusing)

Stochastic part: a (non-normal) distribution

Linear model: $\eta(x) = a + b \cdot x$

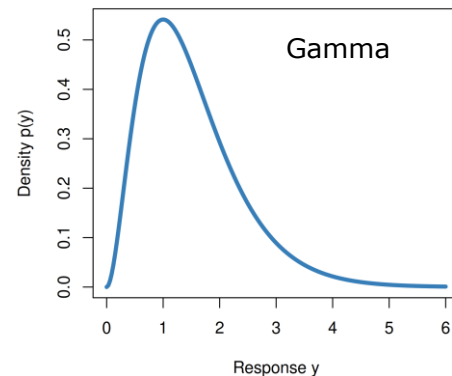
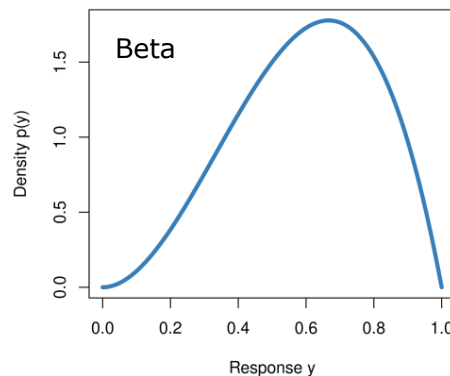
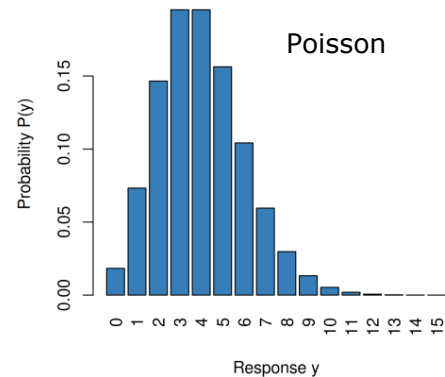
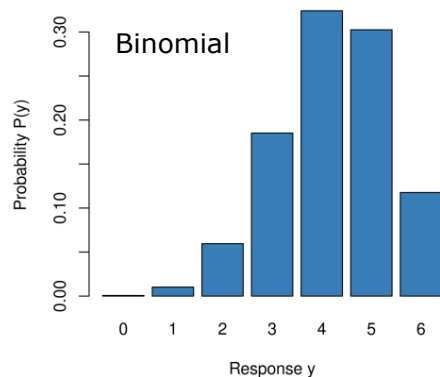
(Inverse) link: $\mu = f(\eta)$

Residual: $y \sim \text{Distribution}(\mu, \dots)$

mean prediction additional scale or dispersion parameters

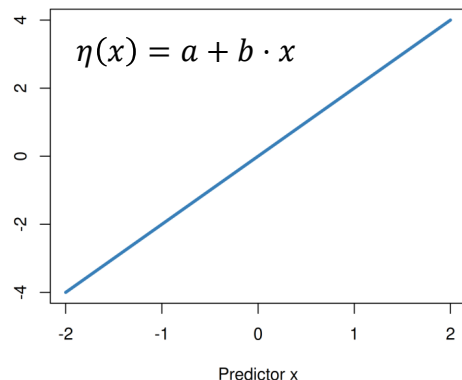
Distribution limited by type of data y :

- discrete or continuous ?
- constrained (lower/upper) or unconstrained ?
- skewed or symmetric ?
- mean-variance relationship ?

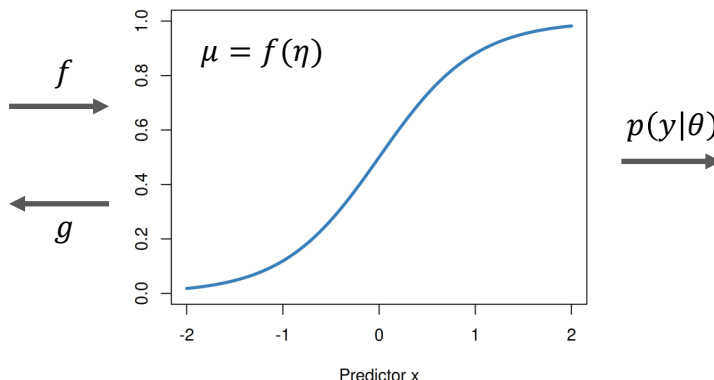


3 parts of the GLM

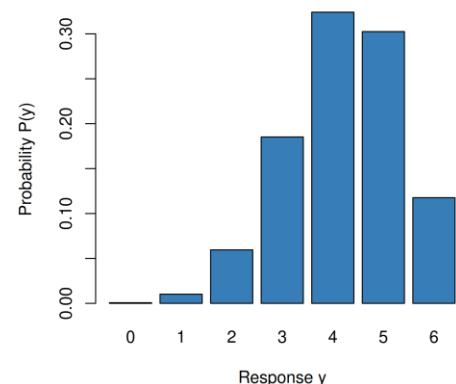
1: linear model



2: (inverse) link



3: residual distribution



- Which predictors ?
- How? Interactions, etc. ?
- All linear models allowed (also ANOVA, ANCOVA, ...)
- Assign **priors** $p(\theta)$ for model parameters !

- Mean prediction $\mu(x)$
- On response scale
- f inverse link function
- g link function

- How is y distributed around mean prediction μ ?
- Defines **likelihood** $p(y|\theta)$

Computation

Logical order of computation

$$\eta = a + b \cdot x$$

$$\mu = f(\eta)$$

$$y \sim \text{Distribution}(\mu)$$

Uses inverse-link f

Pseudo-code for Stan / Nimble / JAGS

```
for(i=1:n){  
  mu[i] = f(a+b*x[i])  
  y[i] ~ Distribution(mu[i])  
}
```

Formula-based notation

$$g(\mu) = a + b \cdot x$$

$$y \sim \text{Distribution}(\mu)$$

Uses link function g

Base-R / rstanarm / brms:

```
glm( y~x, family=Distribution(link=g) )  
  
brm( y~x, family=Distribution(link=g),  
     prior=my.priors )
```

Logistic regression

Logistic regression

Example: Occurrence of a butterfly species
versus temperature

Each observation is a habitat.

Dataset records presence / absence (1/0) of a rare species.

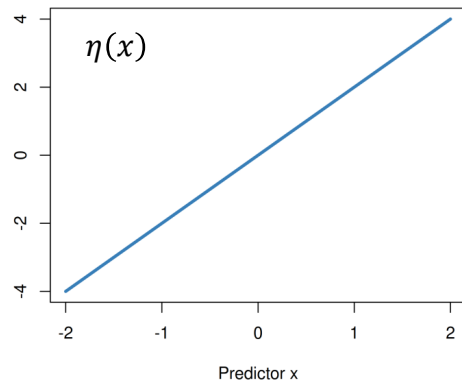
Mean annual temperature of each habitat as predictor.

Q: What is the relationship between temp. and occurrence?
What is the expected probability of occurrence for a new
habitat with a given temperature?



Logistic regression: overview

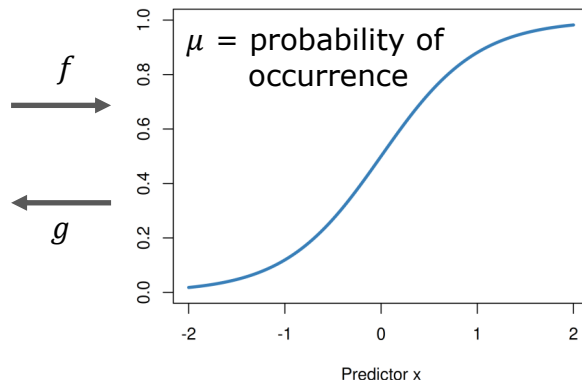
1: linear model



1 predictor temperature:

$$\eta = b_0 + b_1 \cdot \text{temp}$$

2: (inverse) link



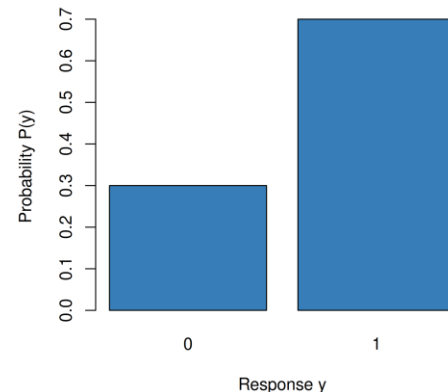
Logit link

$$g = \text{logit}(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$$

Inverse-logit ("logistic")

$$f = \text{logistic}(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)}$$

3: residual distribution



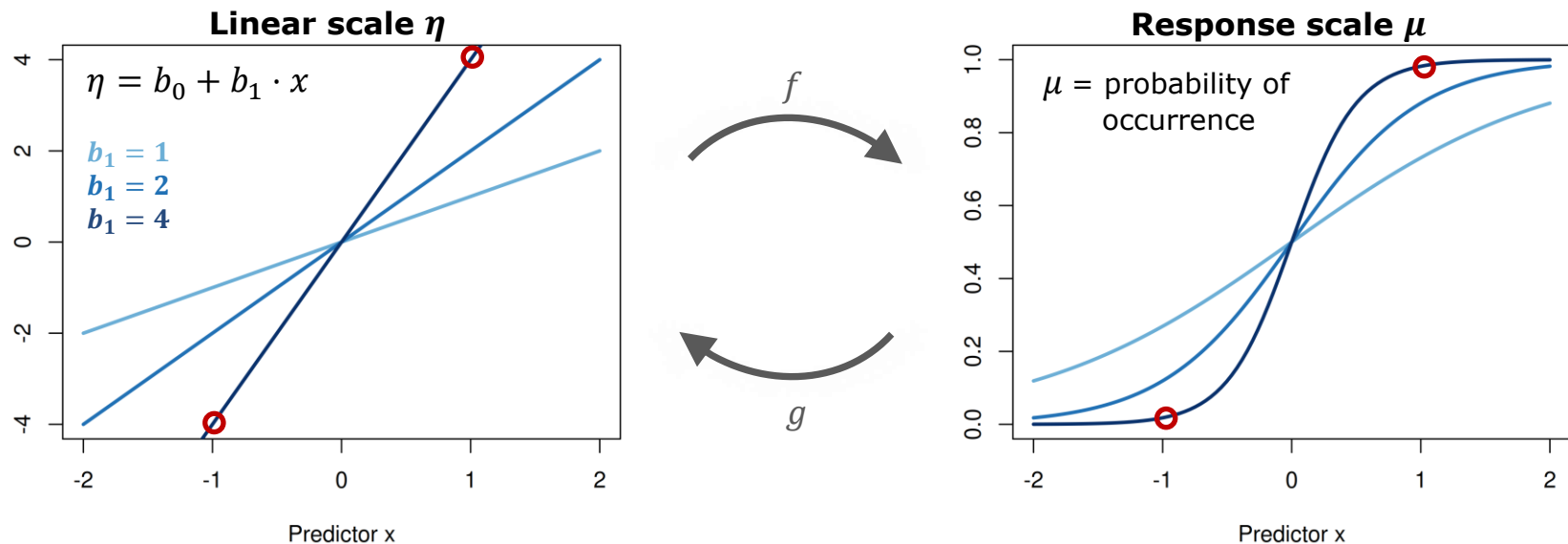
Bernoulli distribution

$$y \sim \text{Bernoulli}(p = \mu)$$

$$= \begin{cases} p & (y = 1) \\ 1 - p & (y = 0) \end{cases}$$

Special case of Binomial distr.
for $N = 1$ trials

Logistic regression: priors



On response scale: $\mu(\eta = +4) = 0.982$ "almost always"
 $\mu(\eta = -4) = 0.018$ "almost never"

→ Scale predictors x & choose meaningful prior for slopes, e.g. $b_1 \sim \text{Normal}(0,1)$

Logistic regression: model fit

Deterministic part
(linear model & link)

$$\text{logit}(p) = b_0 + b_1 \cdot \text{temp}$$

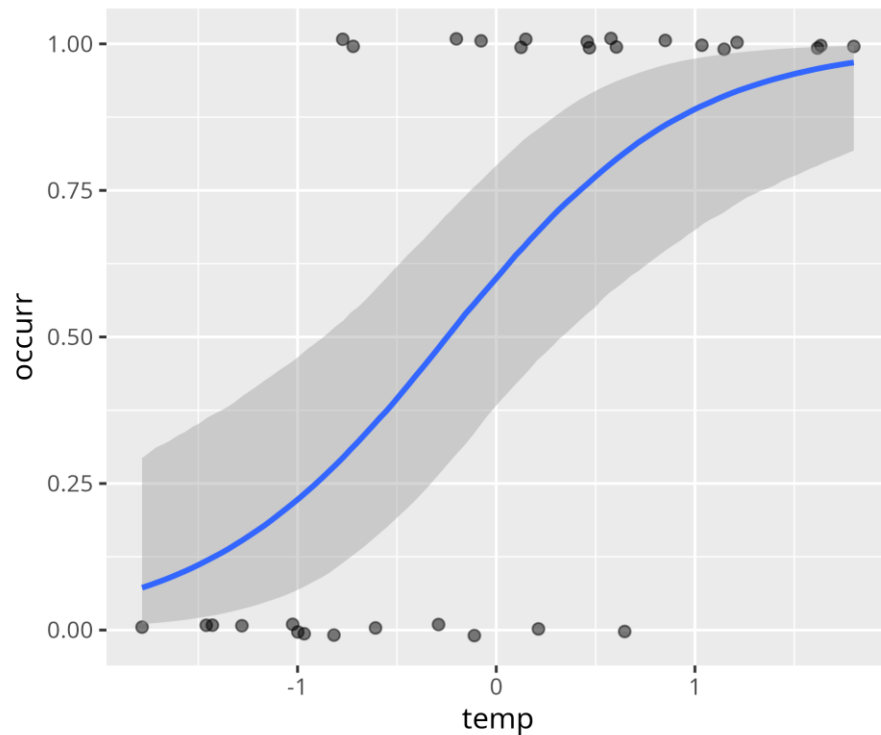
Stochastic part

$$\text{occurr} \sim \text{Bernoulli}(p)$$

Priors

$$b_0 \sim \text{brms-default}$$
$$b_1 \sim \text{Normal}(0,1)$$

```
> brm(occurr~temp, family=bernoulli(link=logit),  
      prior =... )
```



Logistic regression: model fit

```
> brm(occurr~temp, family=bernoulli(link=logit),  
      prior =... )
```

```
Family: bernoulli  
Links: mu = logit  
Formula: occurr ~ temp  
Data: df (Number of observations: 30)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
       total post-warmup draws = 4000
```

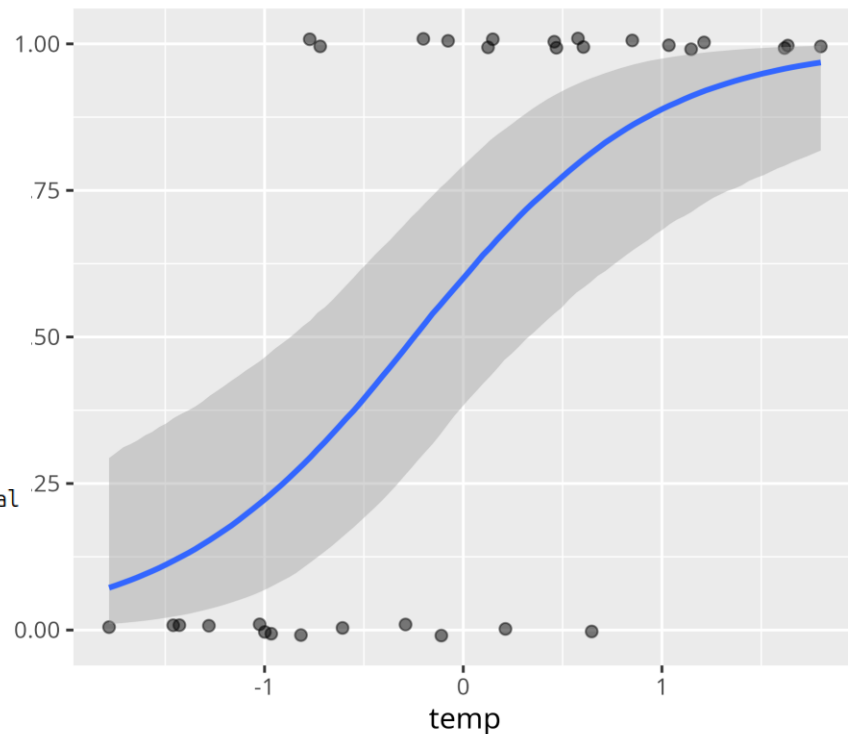
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.41	0.46	-0.48	1.34	1.00	3132	2525
temp	1.70	0.51	0.77	2.78	1.00	2638	2078

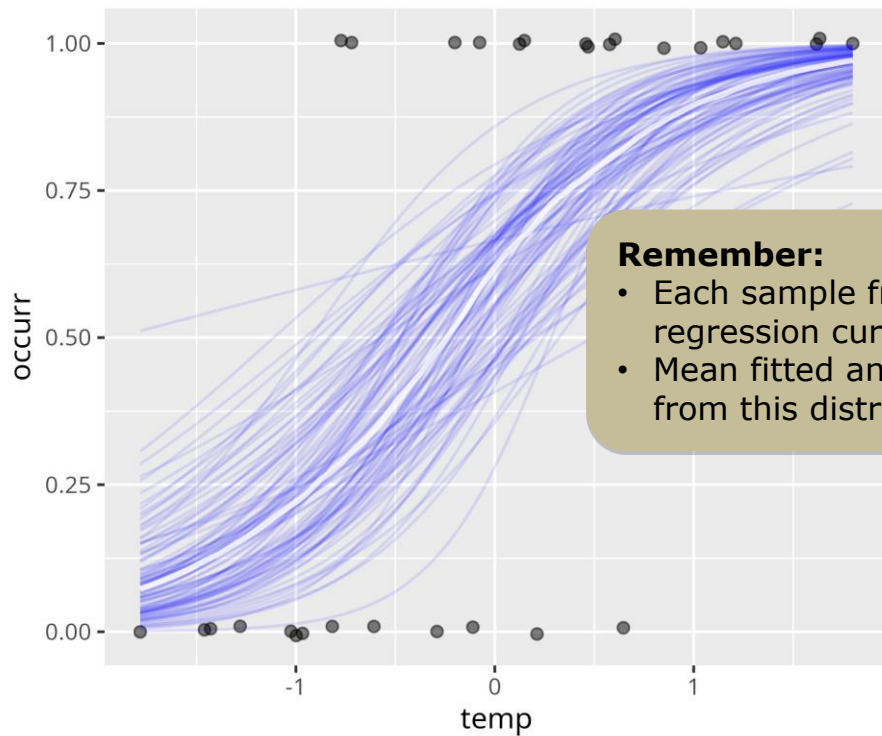
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Q: Is there a positive relationship between temperature and occurrence?

→ Yes, positive: $b_1 = 1.70 [0.77, 2.78]$

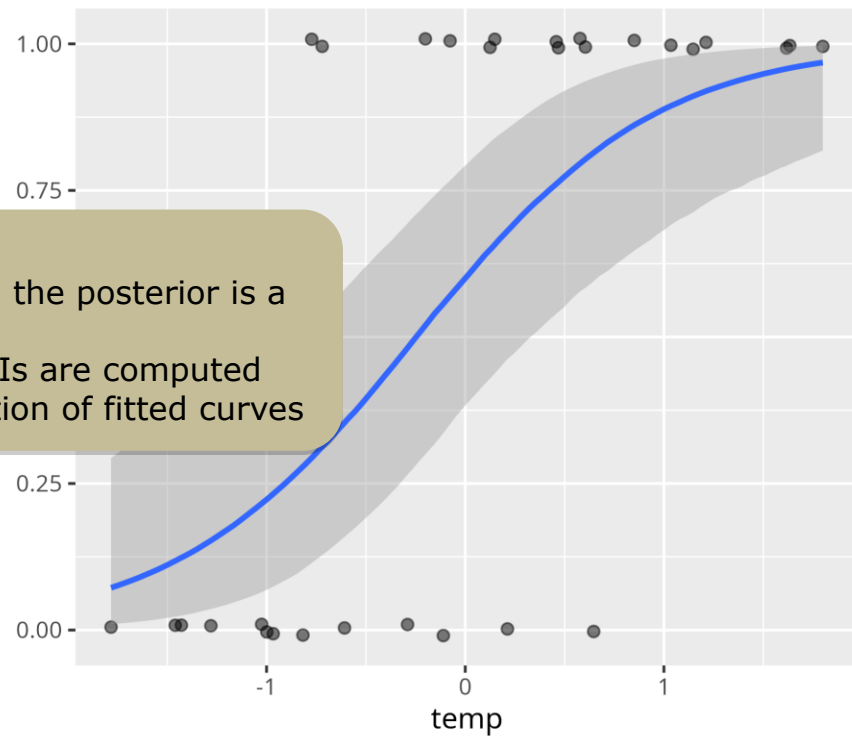


Logistic regression: model fit

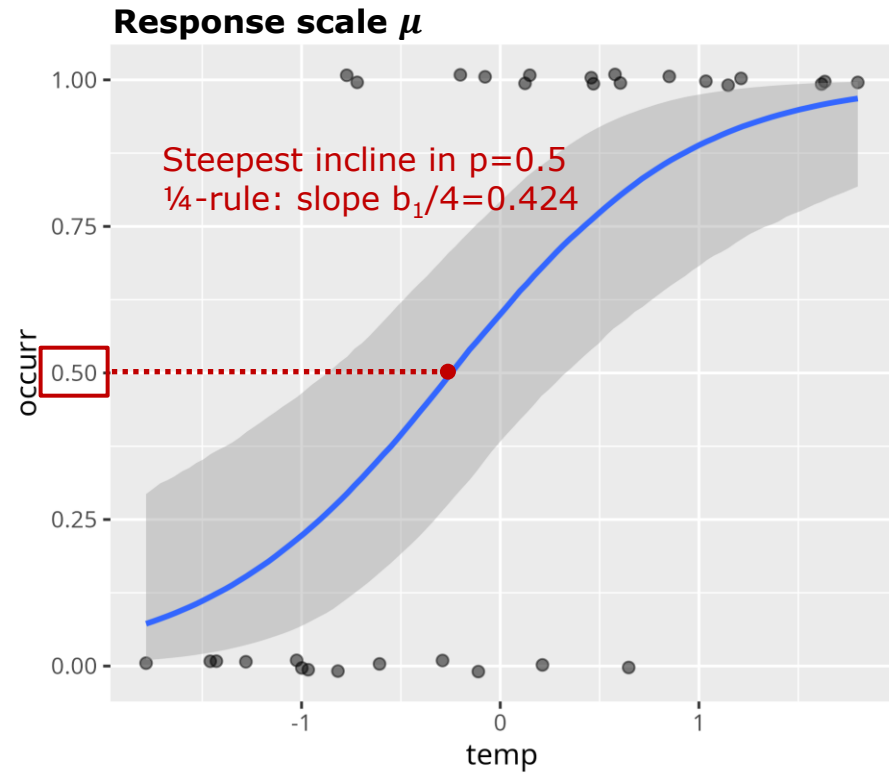
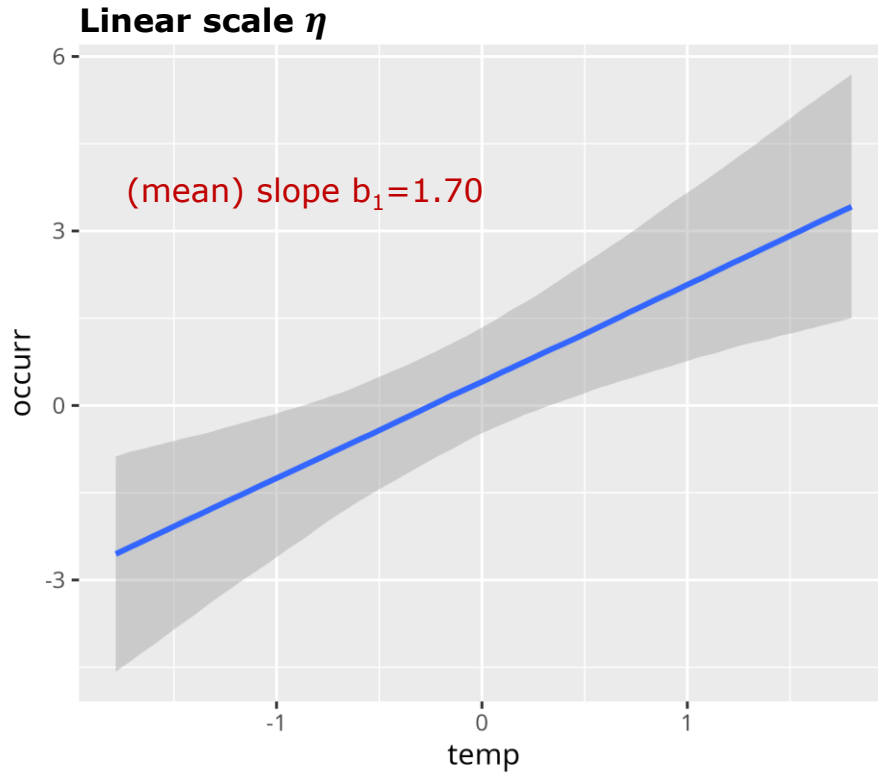


Remember:

- Each sample from the posterior is a regression curve
- Mean fitted and CIs are computed from this distribution of fitted curves

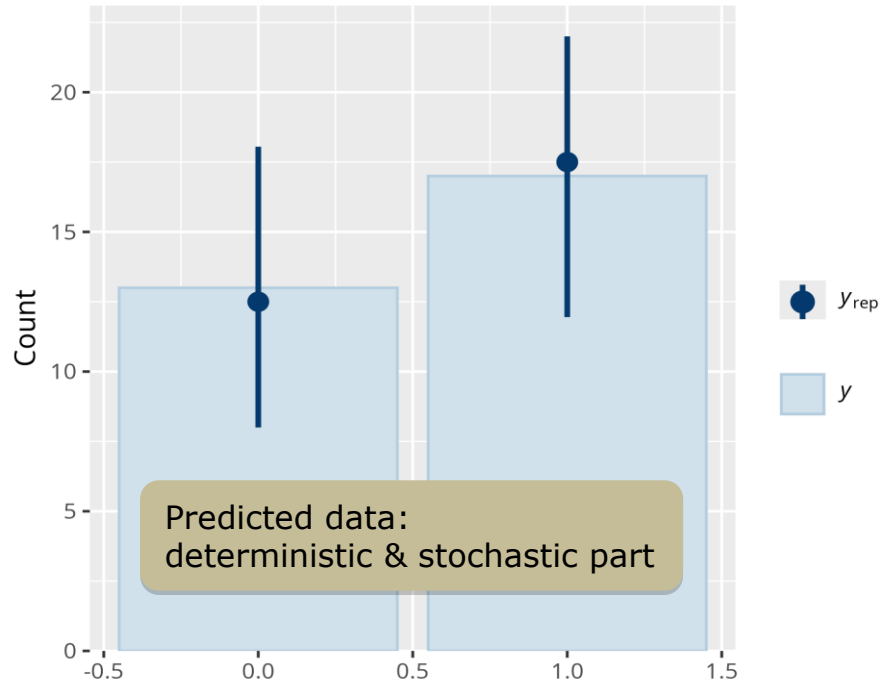


Logistic regression: model fit

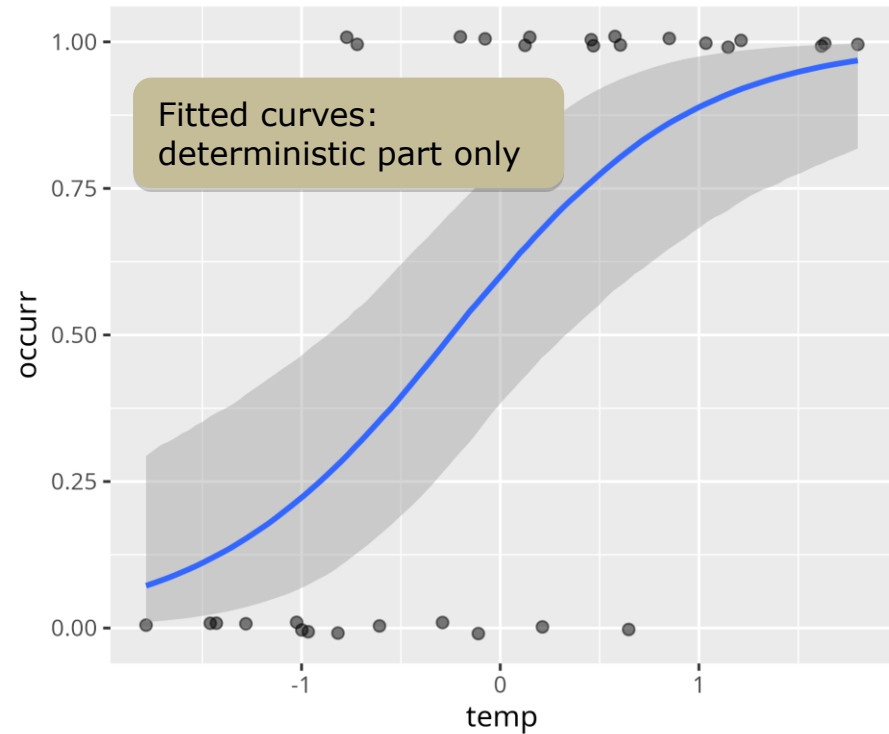


Logistic regression: evaluation

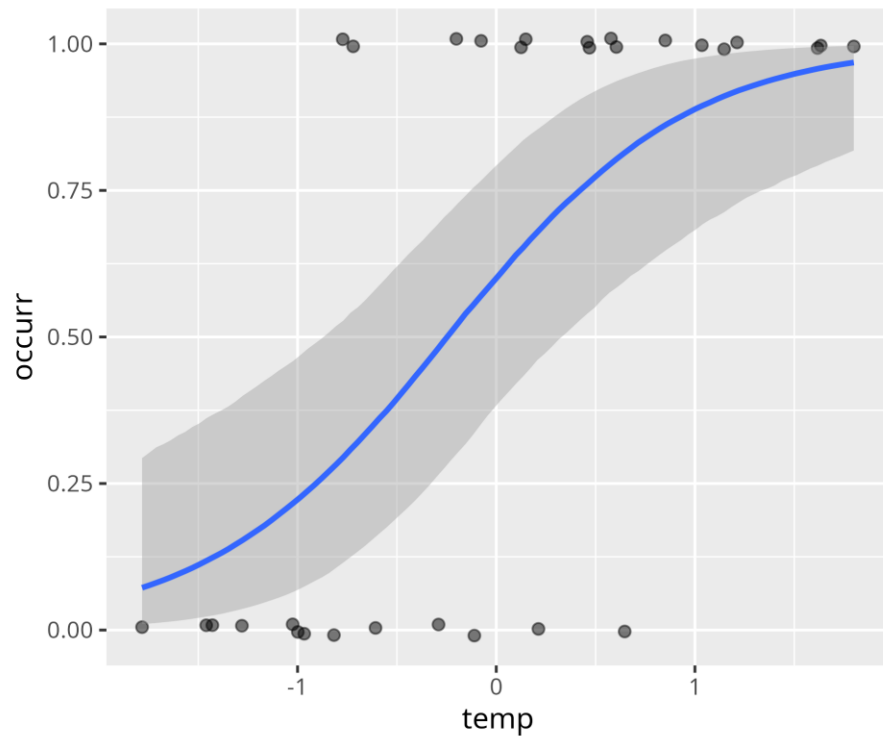
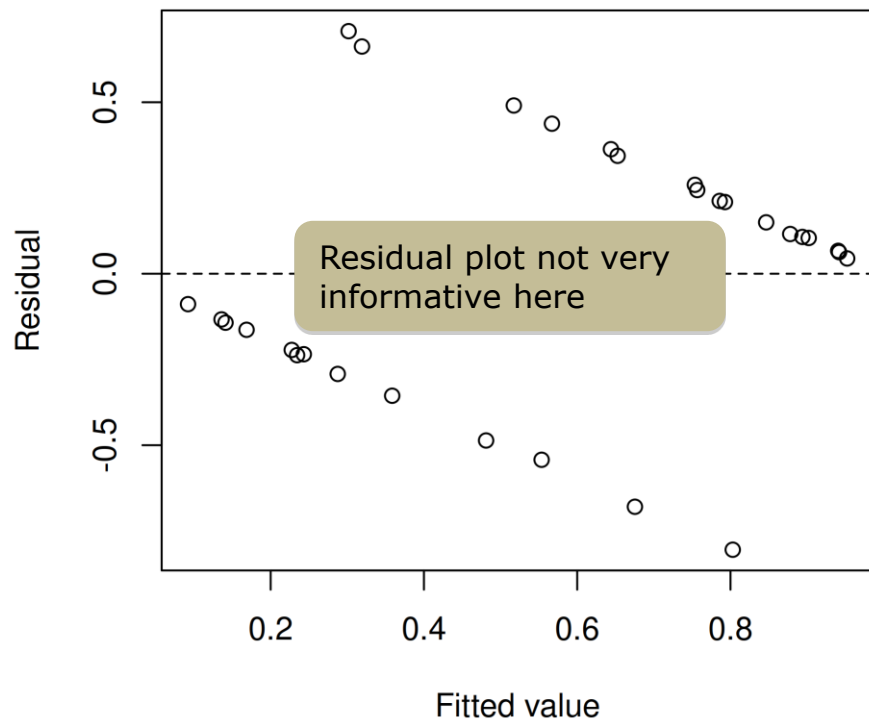
```
> pp_check(fit, type="bars")
```



```
> plot(conditional_effects(fit,...))
```

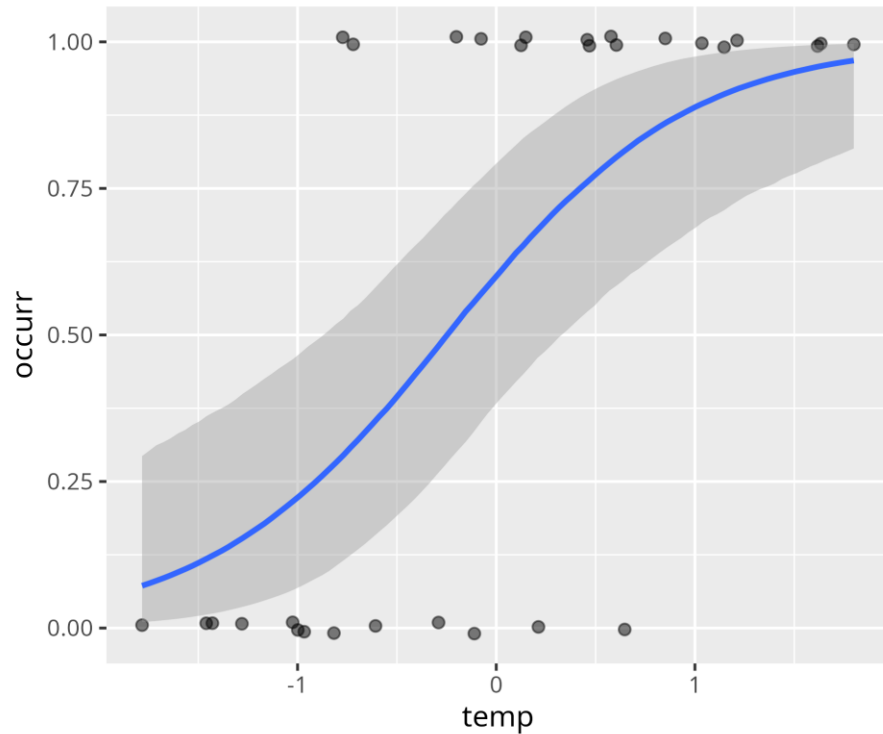
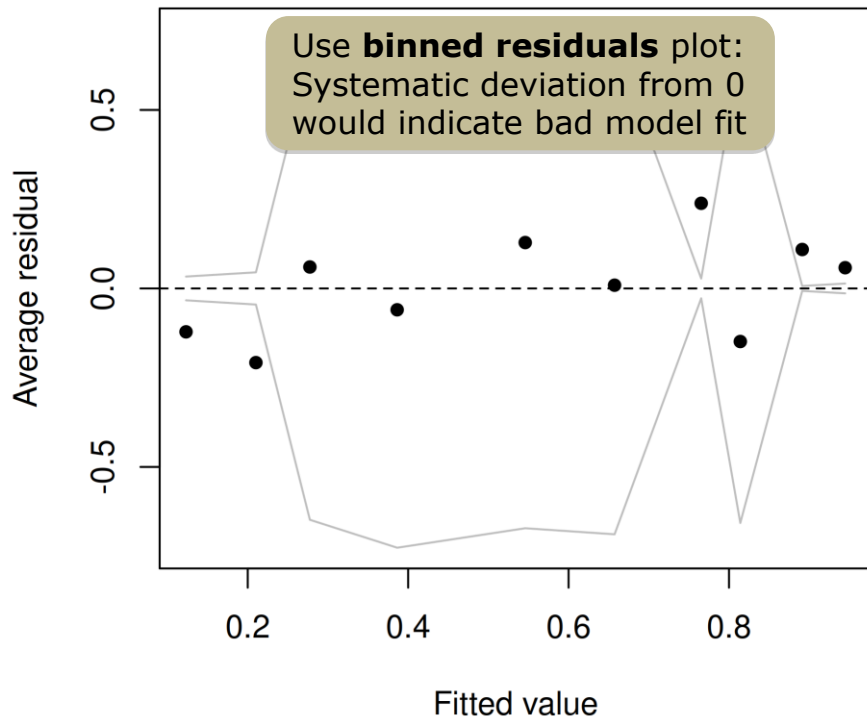


Logistic regression: evaluation



Logistic regression: evaluation

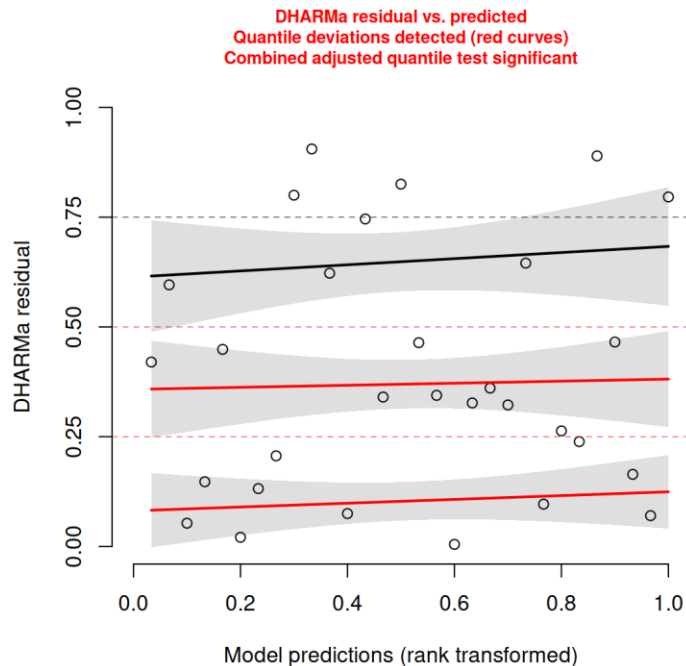
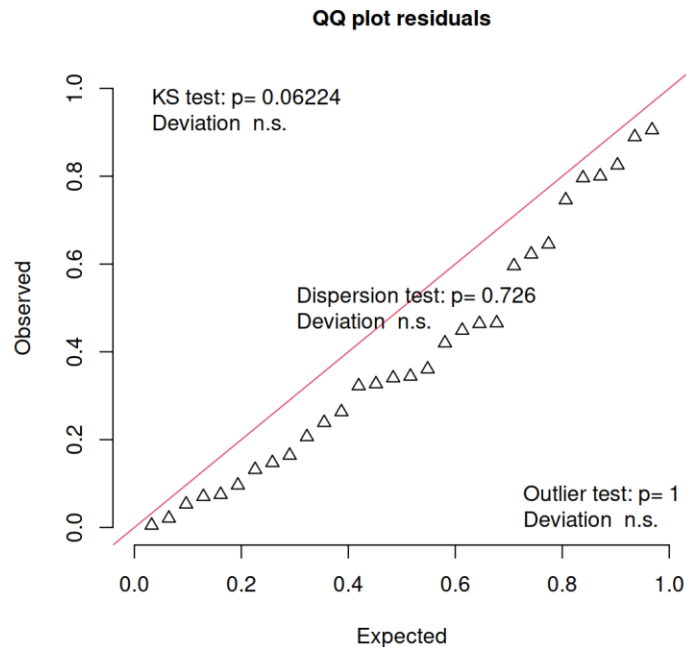
```
> arm::binnedplot(fitted, residuals)
```



Logistic regression: evaluation

DHARMa package for GLM(M):
Checks residuals, dispersion, outliers

```
> DHARMa.helpers::dh_check_brms(fit)
```



Logistic regression (Niche model)

Example: Occurrence of a butterfly species
versus temperature

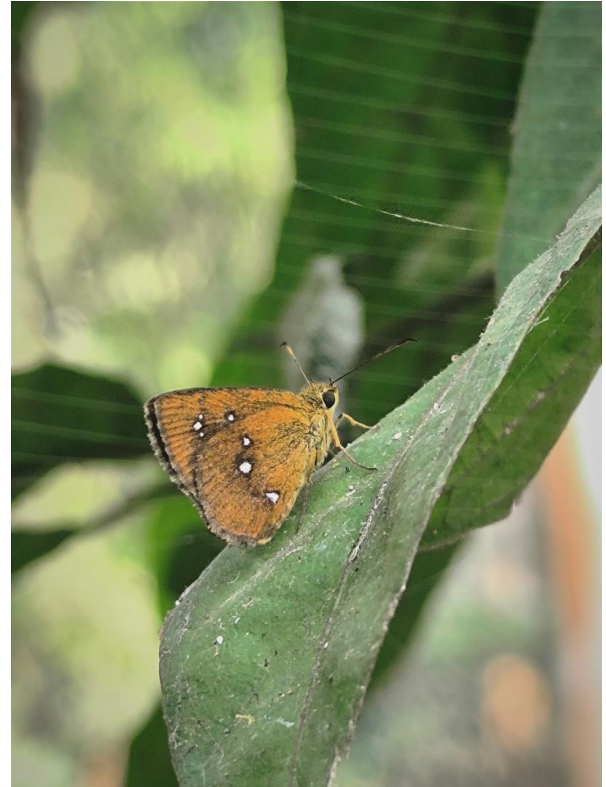
Each observation is a habitat.

Dataset records presence / absence (1/0) of a rare species.

Mean annual temperature of each habitat as predictor.

New data collected in warmer habitats.

Q: What is the optimal temperature for this species ?



Logistic regression: model fit

Deterministic part

(linear model & link)

$$\text{logit}(p) = b_0 + b_1 \cdot \text{temp} + b_2 \cdot \text{temp}^2$$

Stochastic part

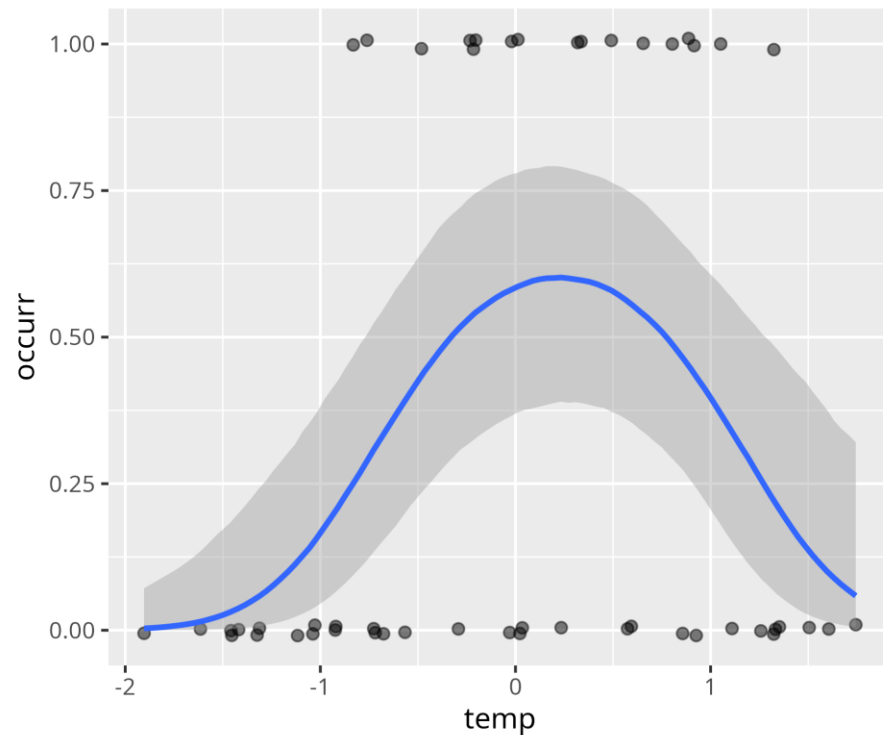
$$\text{occurr} \sim \text{Bernoulli}(p)$$

Priors

$$b_0 \sim \text{brms-default}$$

$$b_1 \sim \text{Normal}(0,1)$$

```
> brm(occurr ~ temp + I(temp^2),  
      family = bernoulli(link=logit),  
      prior = ... )
```



Logistic regression: model fit

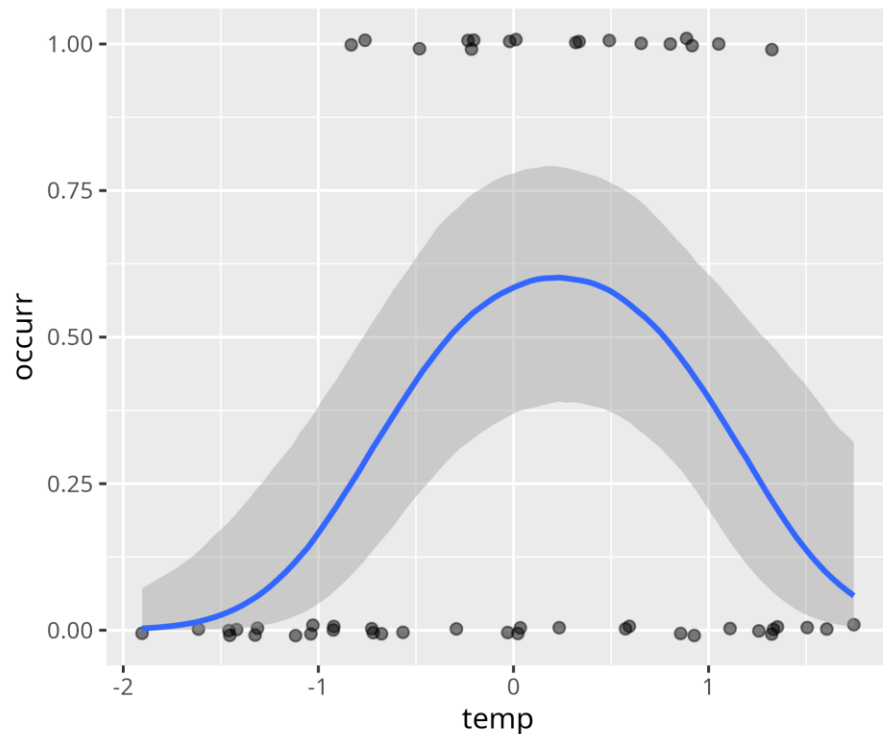
```
> brm(occurr ~ temp + I(temp^2),  
      family = bernoulli(link=logit),  
      prior = ... )
```

```
Family: bernoulli  
Links: mu = logit  
Formula: occurr ~ temp + I(temp^2)  
Data: df (Number of observations: 50)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
       total post-warmup draws = 4000
```

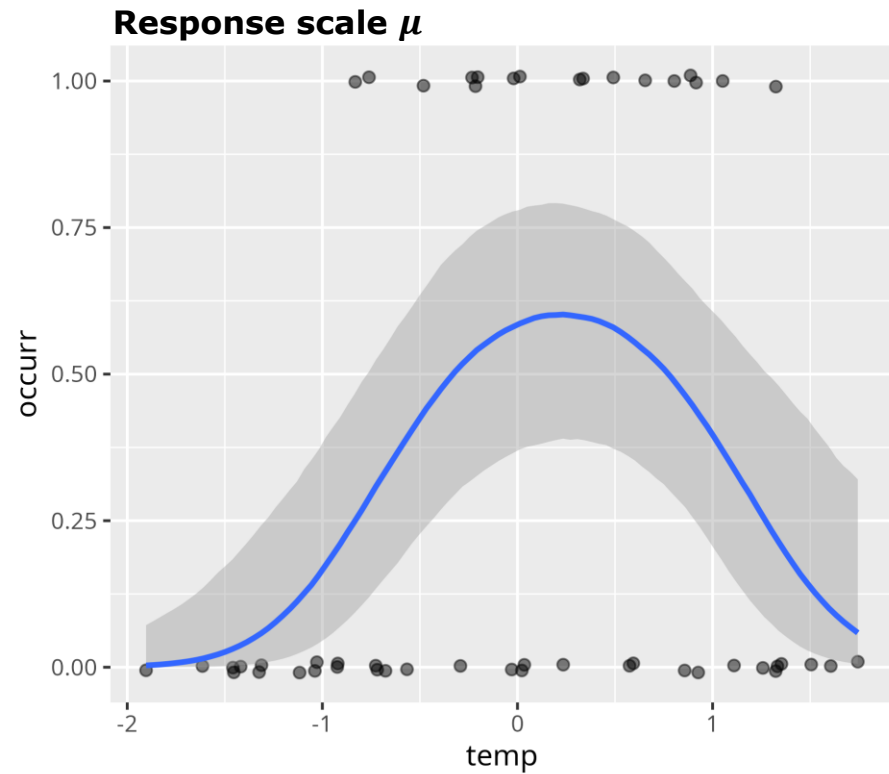
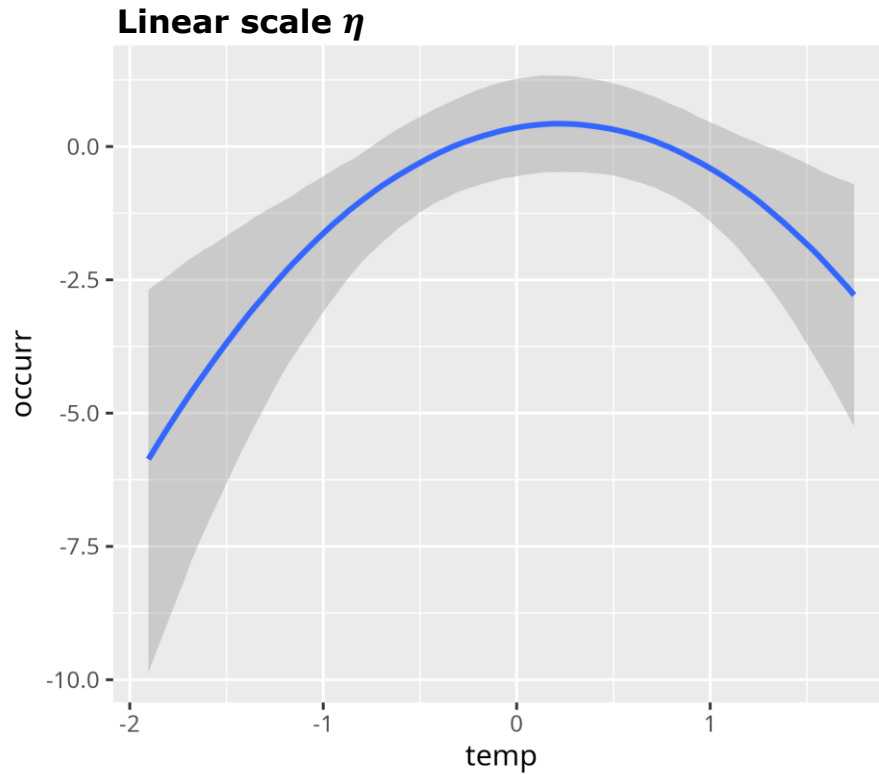
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.36	0.46	-0.55	1.26	1.00	5301	3161
temp	0.63	0.42	-0.15	1.49	1.00	2167	2154
ItempE2	-1.41	0.46	-2.36	-0.53	1.00	2031	2262

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).



Logistic regression: model fit



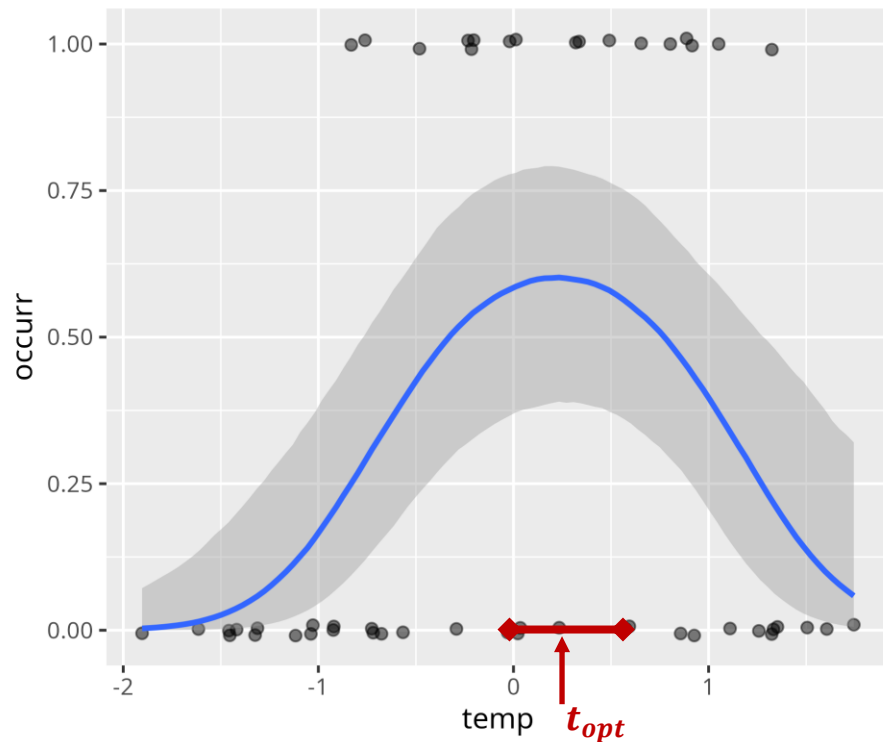
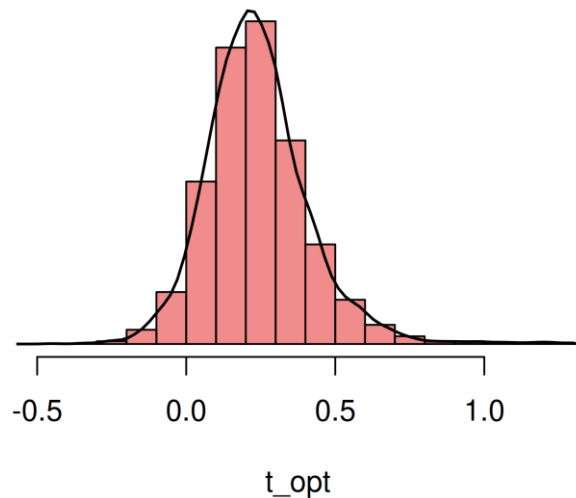
Logistic regression: model fit

Q: What is the optimal temperature?
(peak of the curve)

some calculus $\rightarrow t_{\text{opt}} = -\frac{b_1}{2b_2}$

compute t_{opt} for each posterior sample

$\rightarrow$ mean 0.22, 90% CI [0.01, 0.54]



Binomial regression

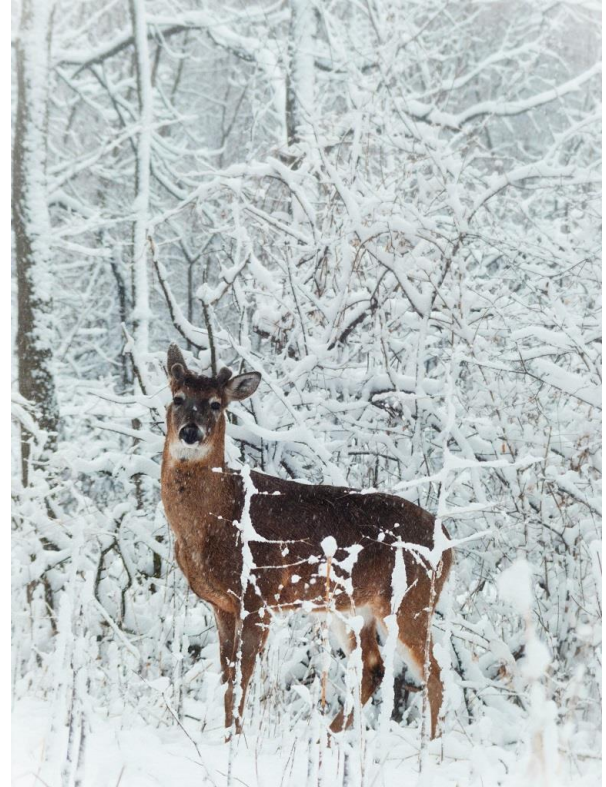
Example: Deer survival probability
versus vegetation and predation

Monitored deer populations several habitats.

Recorded number of deer before and after winter.

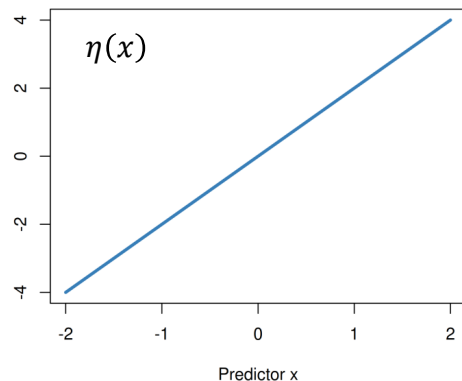
Vegetation index (NDVI) and predator presence (yes/no).

Q: How much does vegetation affect survival probability?
Must control for predation. Is there an interaction?

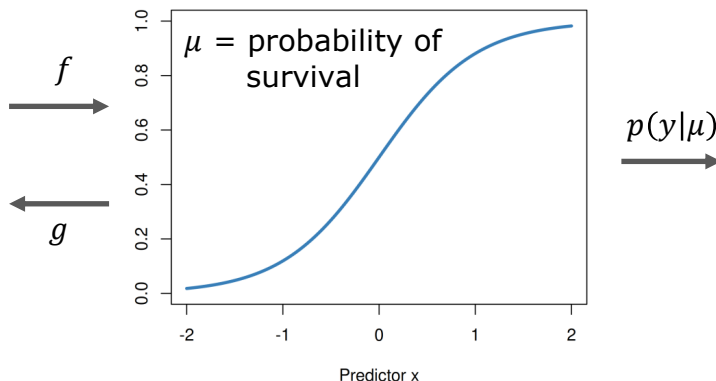


Binomial regression: overview

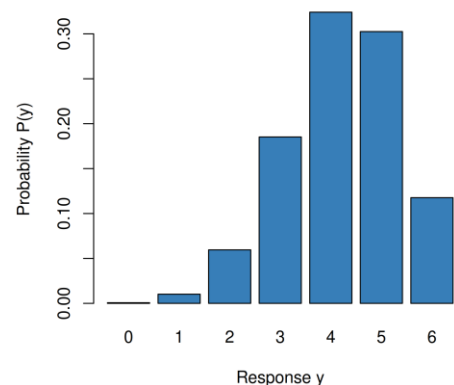
1: linear model



2: (inverse) link



3: residual distribution



Logit link

$$g = \text{logit}(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$$

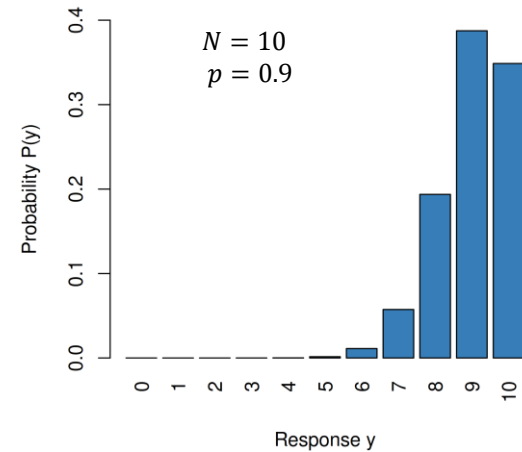
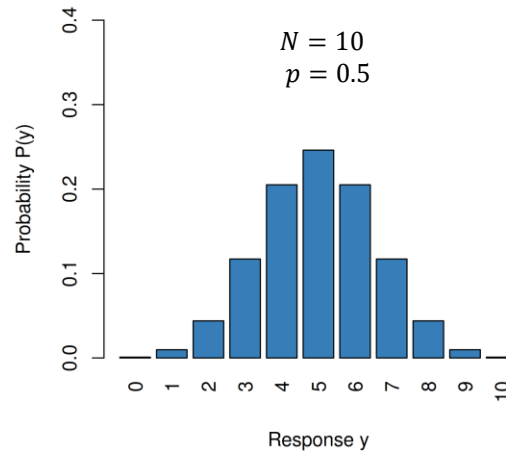
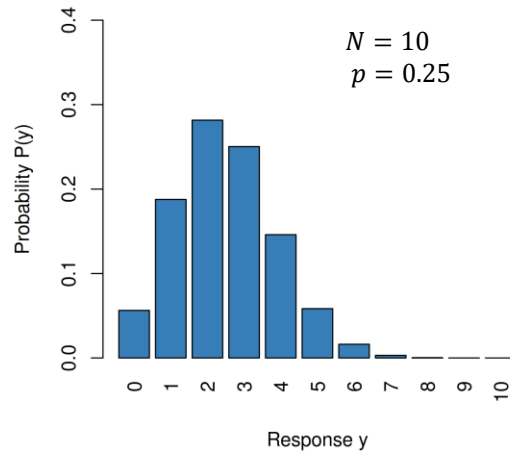
Inverse-logit ("logistic")

$$f = \text{logistic}(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)}$$

Binomial distribution

$$y \sim \text{Binomial}(N, p = \mu)$$

Binomial distribution



Discrete distribution, bounded between 0 and N

Mean: $N \cdot p$

Sdev: $\sqrt{Np(1-p)}$ (increases with N)

Every datapoint y_i can have its own N_i

Binomial regression: ANCOVA model

Deterministic part

(linear model & link)

$$\text{logit}(p) = a_0 + a_1 \cdot x_{pred} + (b_0 + b_1 \cdot x_{pred}) \cdot \text{vegetation}$$

Dummy-coding

$$x_{pred} = \begin{cases} 0, & \text{predator} = \text{no} \\ 1, & \text{predator} = \text{yes} \end{cases}$$

Stochastic part

$$\text{survived} \sim \text{Binomial}(\text{total}, p)$$

Priors

$$a_0 \sim \text{brms-default}$$

$$a_1, b_0, b_1 \sim \text{Normal}(0, 1)$$

```
> brm(survived | trials(total) ~ vegetation * predator,  
      family = binomial(link=logit),  
      prior = ... )
```

	survived	total	vegetation	predator
1	7	40	0.2663940	1
2	21	25	0.7321366	0
3	5	15	0.6159937	1
4	3	37	0.2512467	1
5	26	28	0.9494554	1
6	18	20	0.9491275	0
7	4	14	0.2162431	0
8	26	27	0.8501039	0
9	26	35	0.5212167	0
10	6	11	0.5949854	0
11	14	16	0.5974067	0
12	4	13	0.3150053	1
13	33	39	0.7844620	1
14	2	22	0.2627381	1
15	4	20	0.4647540	1

Binomial regression: model fit additive

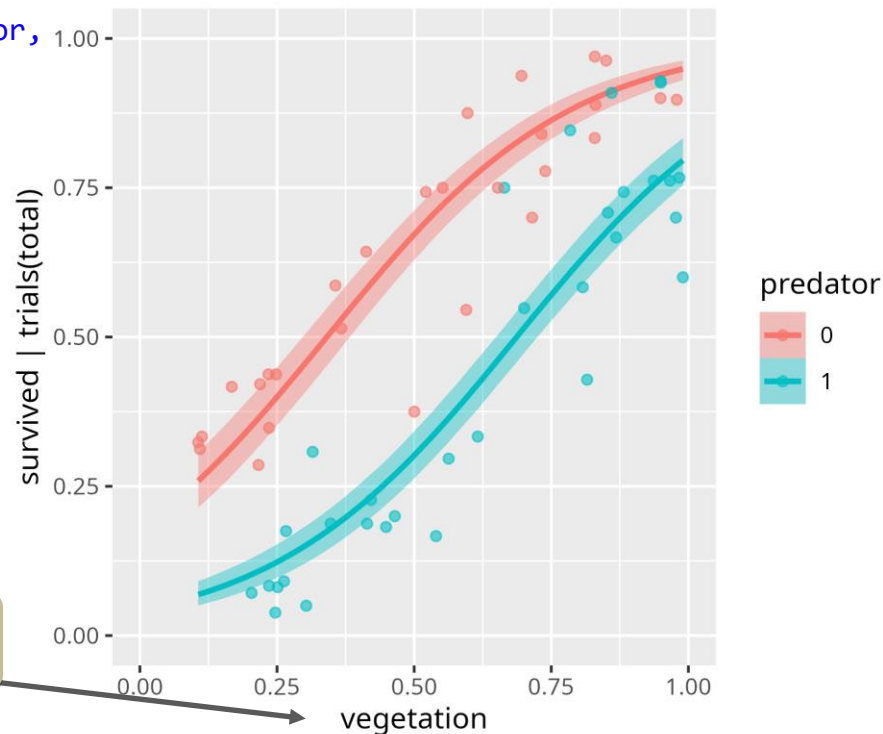
```
> brm(survived|trials(total) ~ vegetation+predator,  
      family = binomial(link=logit),  
      prior = ... )
```

Family: binomial
Links: mu = logit
Formula: survived | trials(total) ~ scale(vegetation) + predator
Data: df (Number of observations: 60)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Regression Coefficients:

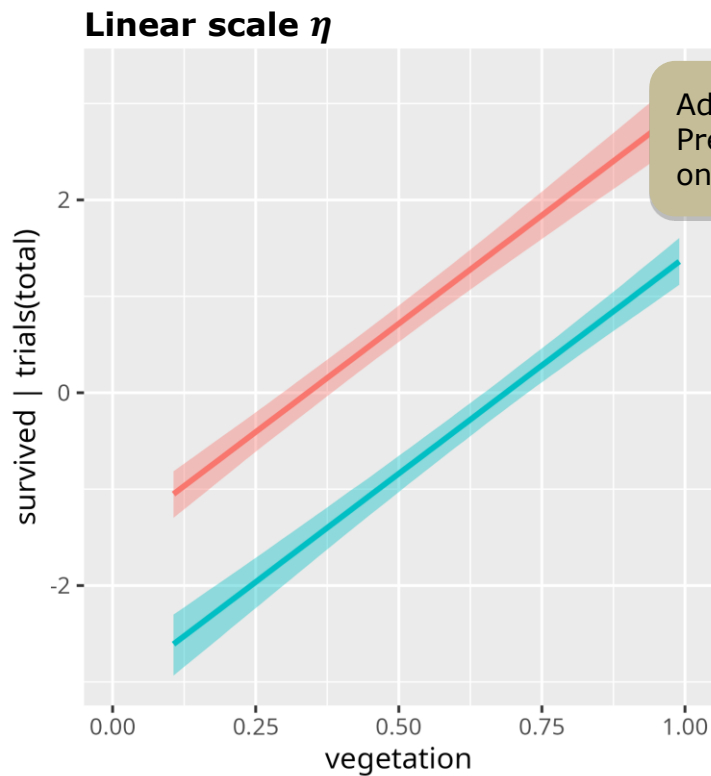
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.03	0.10	0.84	1.23	1.00	3104	3066
scalevegetation	1.27	0.07	1.13	1.41	1.00	3207	2747
predator1	-1.56	0.14	-1.83	-1.29	1.00	3046	2680

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

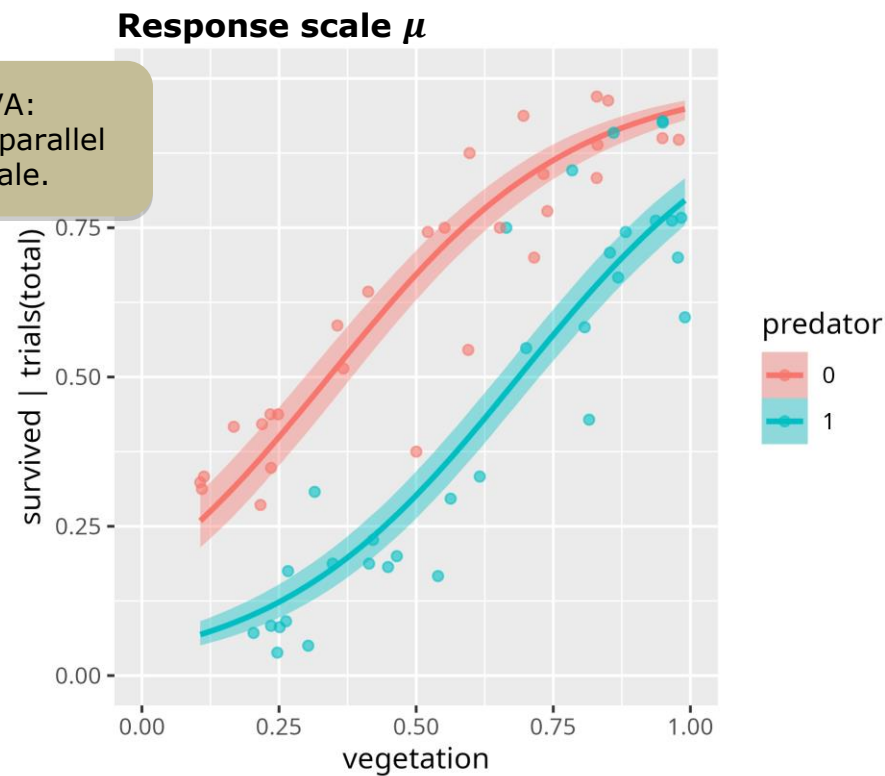


Can also use scale(...) in the model formula.
Predictions can be generated on original scale

Binomial regression: model fit additive



Additive ANCOVA:
Predictions are parallel
on the linear scale.



Binomial regression: model fit interaction

```
> brm(survived|trials(total) ~ vegetation*predator,  
      family = binomial(link=logit),  
      prior = ... )
```

```
Family: binomial  
Links: mu = logit  
Formula: survived | trials(total) ~ scale(vegetation) * predator  
Data: df (Number of observations: 60)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
       total post-warmup draws = 4000
```

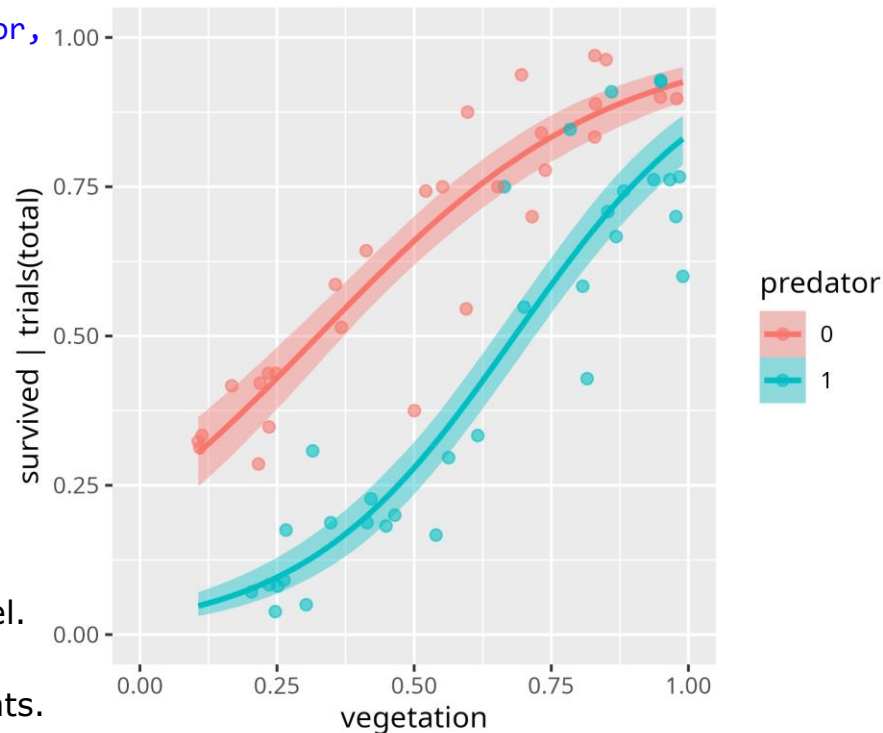
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.93	0.10	0.73	1.14	1.00	2764
scalevegetation	1.07	0.10	0.88	1.27	1.00	2284
predator1	-1.51	0.14	-1.81	-1.23	1.00	3140
scalevegetation:predator1	0.39	0.14	0.12	0.67	1.00	2125

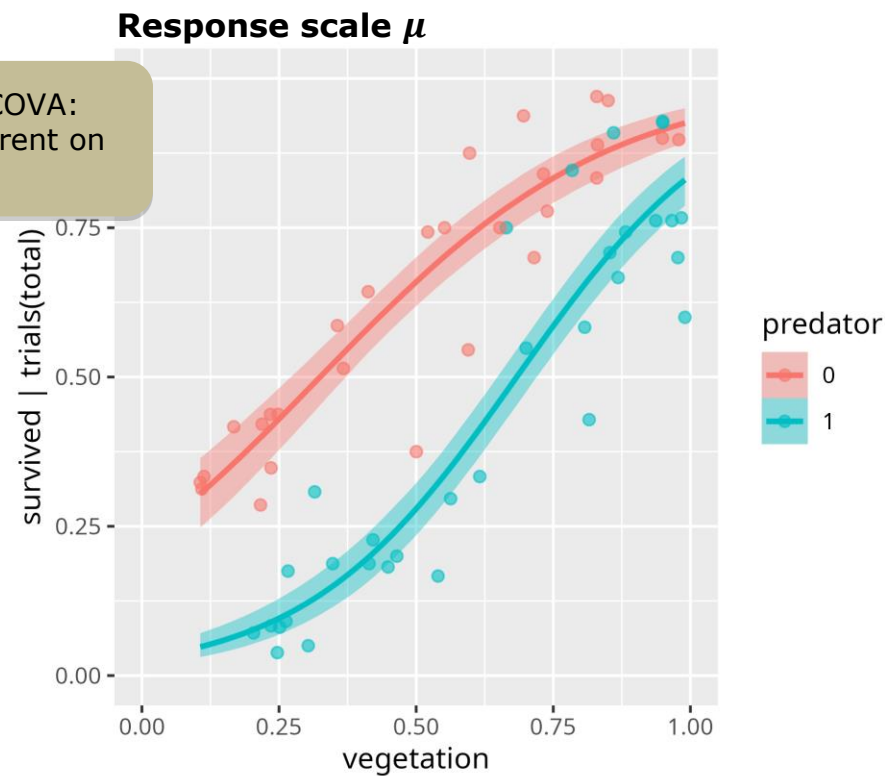
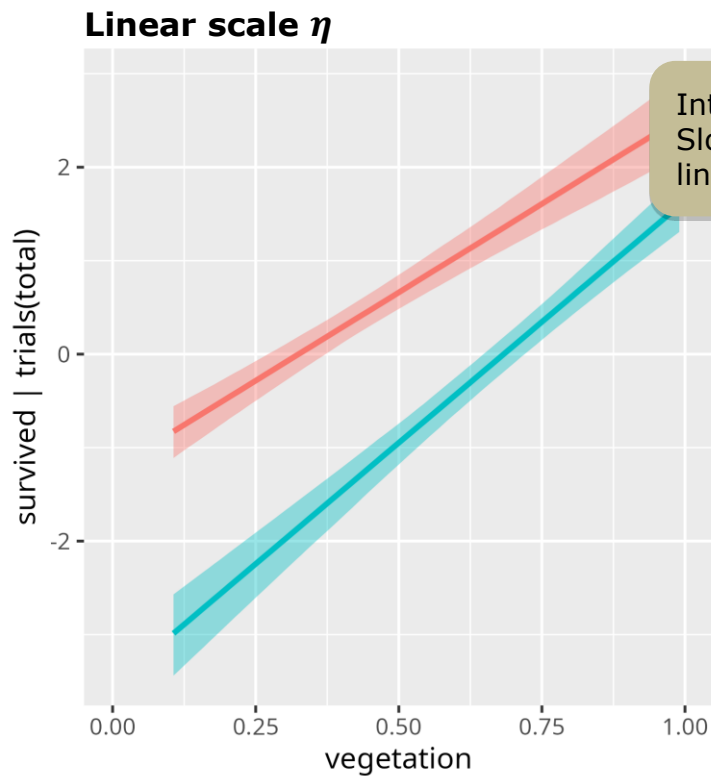
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Alternative: model comparison against additive model.

→ Vegetation has a stronger effect in predator habitats.
Important as shelter to hide from predators.

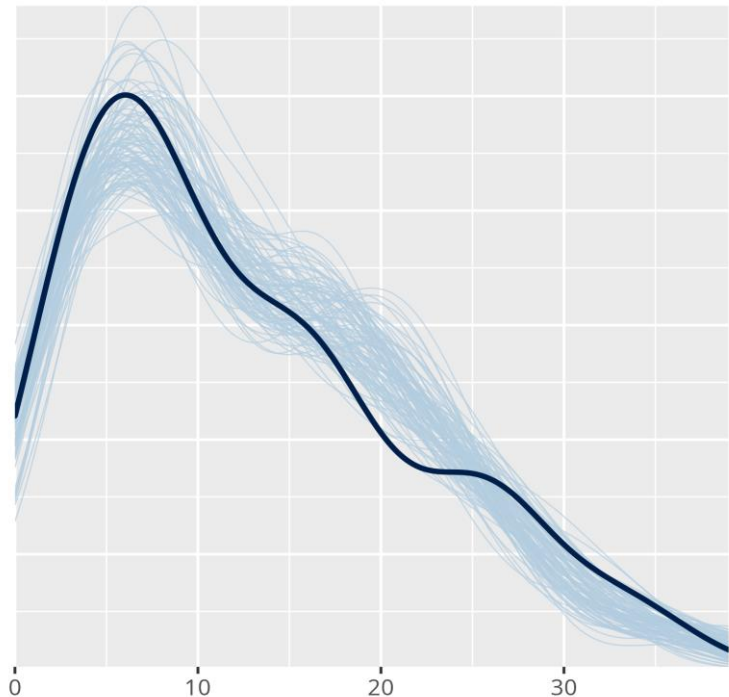


Binomial regression: model fit interaction

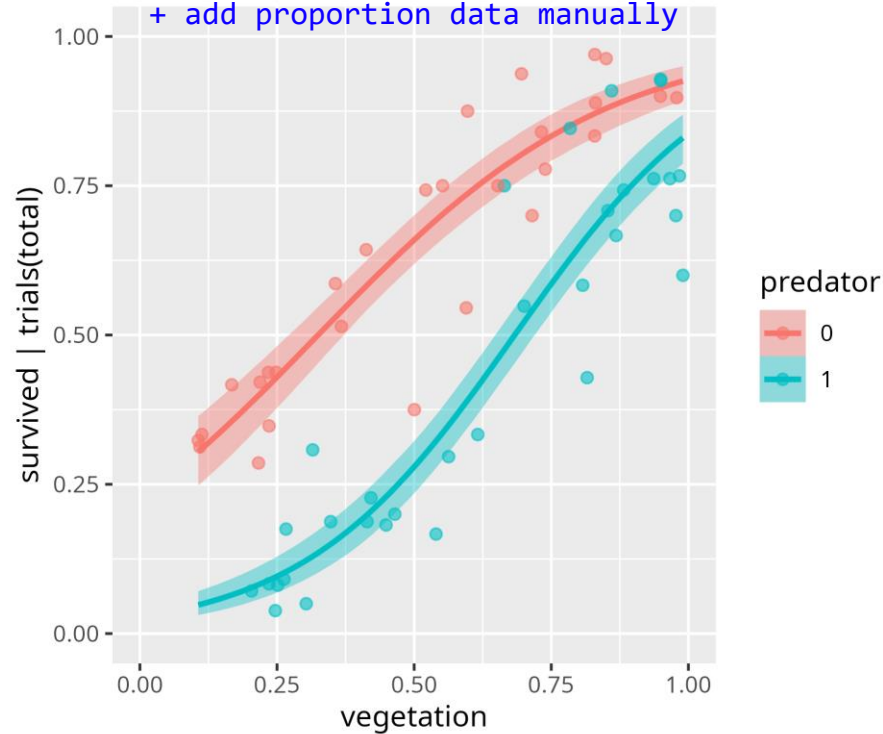


Binomial regression: evaluation

```
> pp_check(fit, ndraws=100)
```

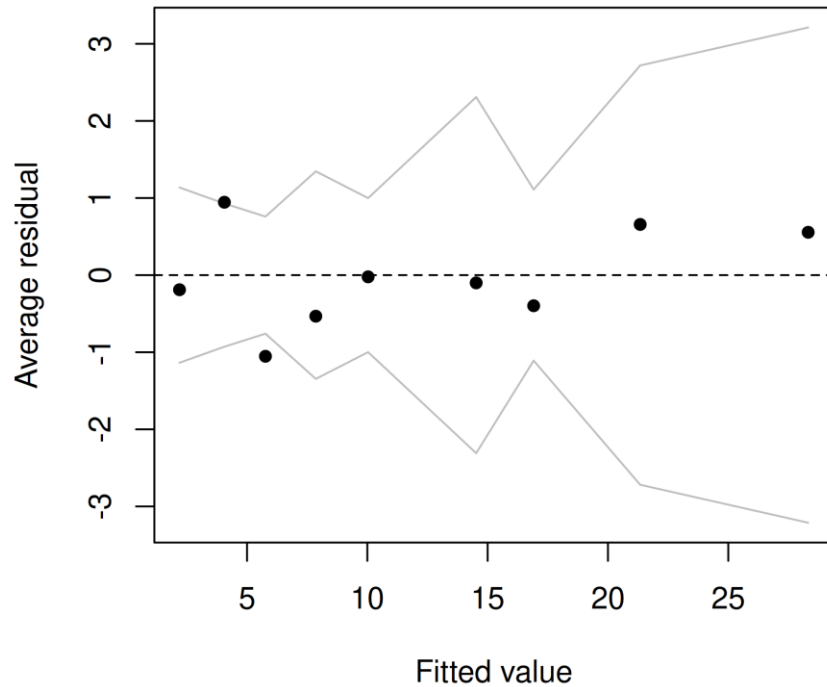


```
> plot(conditional_effects(fit,...))  
+ add_proportion_data_manually
```

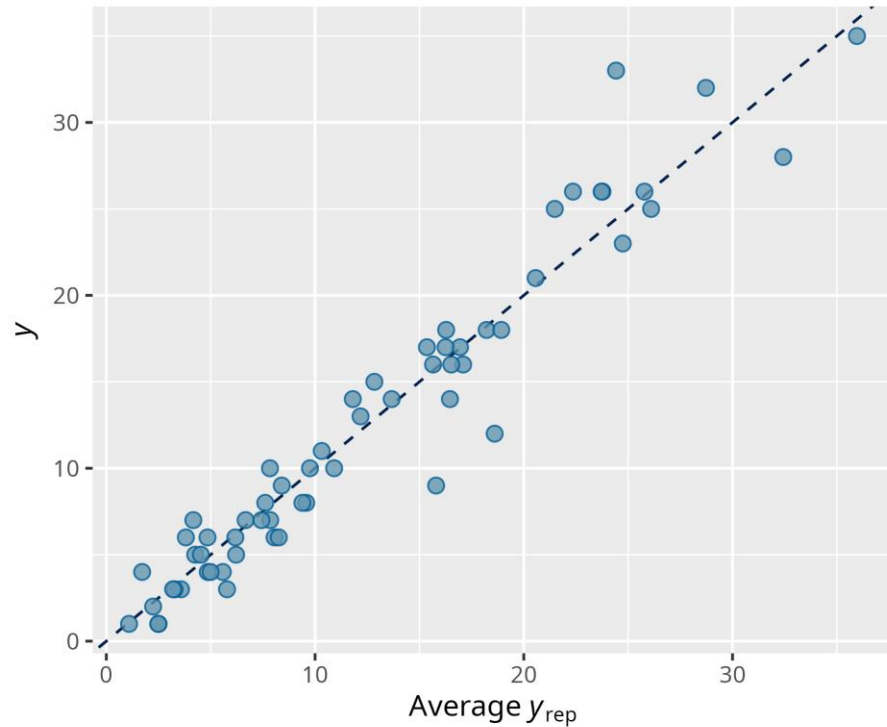


Binomial regression: evaluation

```
> arm::binnedplot(fitted, residuals)
```

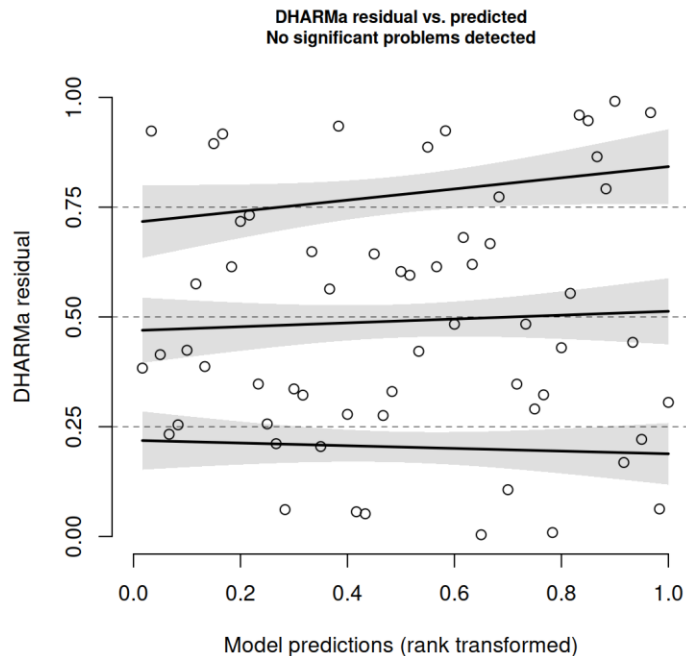
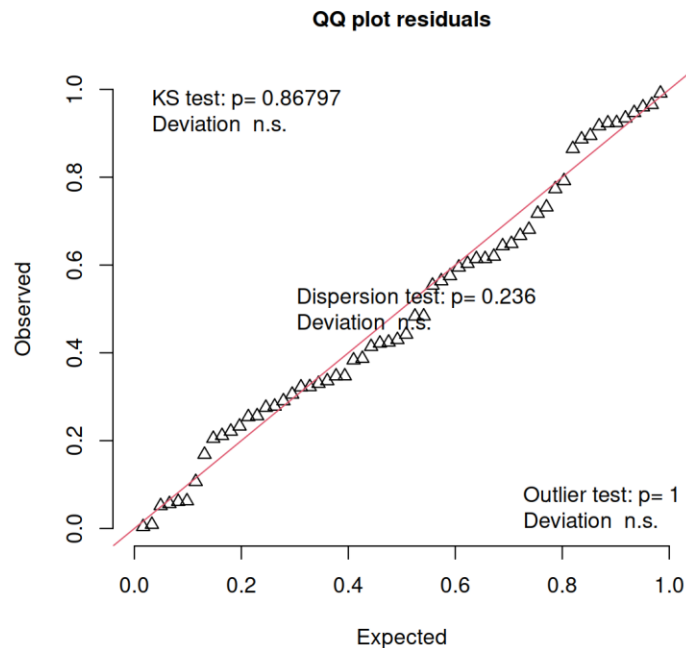


```
> pp_check(fit, type=„scatter_avg“)
```



Binomial regression: evaluation

```
> DHARMA.helpers::dh_check_brms(fit)
```



*Continuous example:
Beta regression*

Beta regression

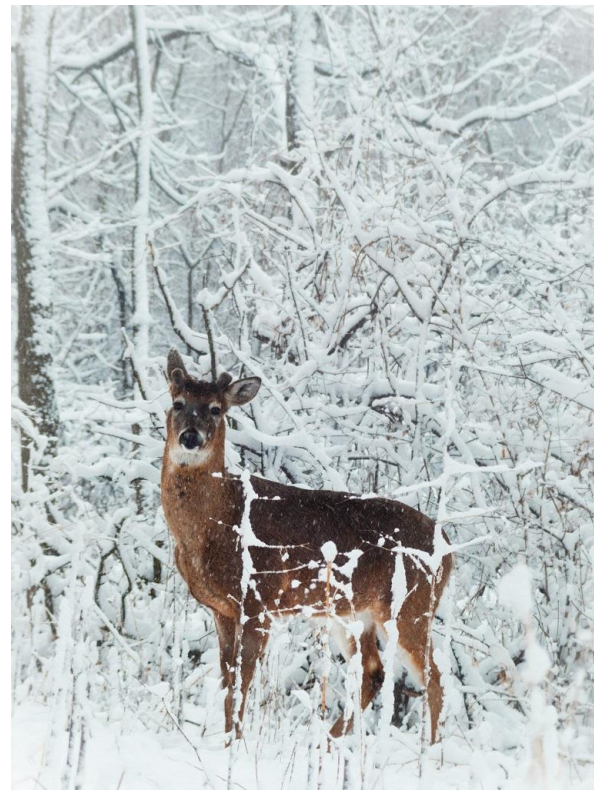
Example: Deer survival probability
versus vegetation and predation

Same data as in Binomial regression.

But this time it's an old dataset where they only recorded
the **ratio** $\in [0, 1]$ of surviving individuals, not actual numbers

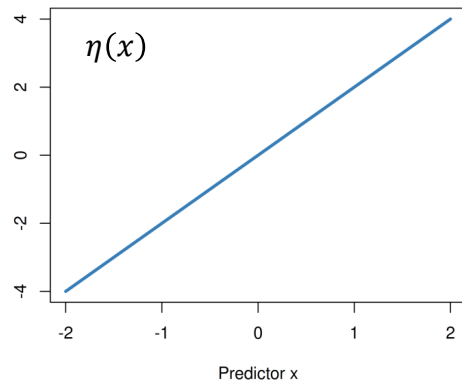
Bad practice: e.g. 50/100 contains more information than 1/2.
When transforming both to 0.5, this information is lost.

Q: How much does vegetation affect survival probability?
Must control for predation.

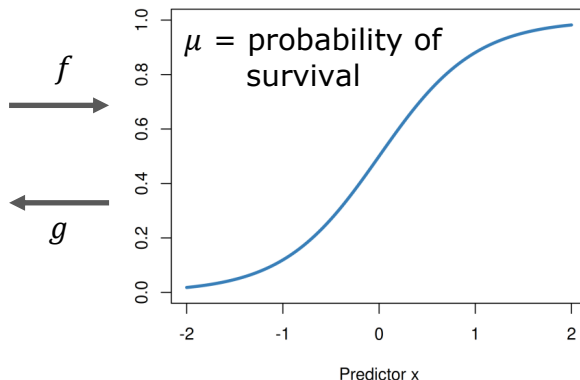


Beta regression: overview

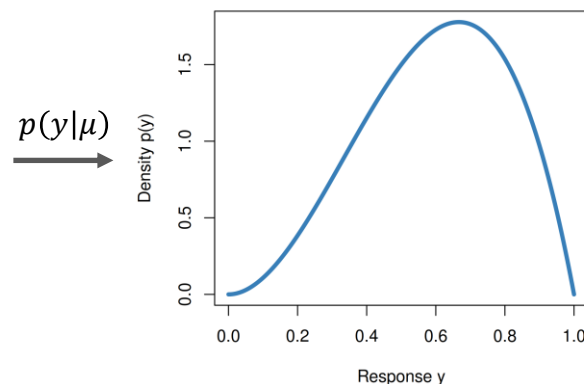
1: linear model



2: (inverse) link



3: residual distribution



Logit link

$$g = \text{logit}(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$$

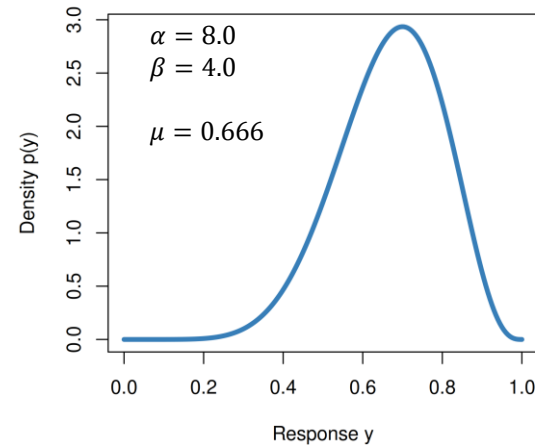
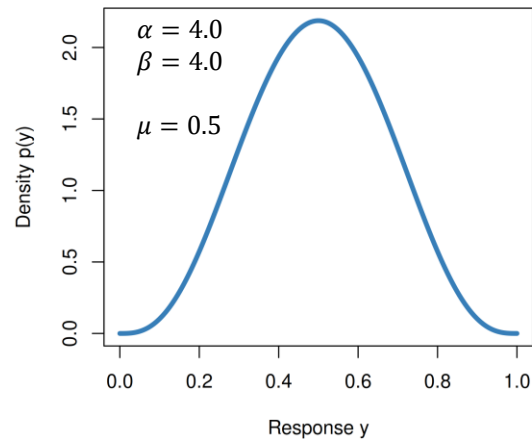
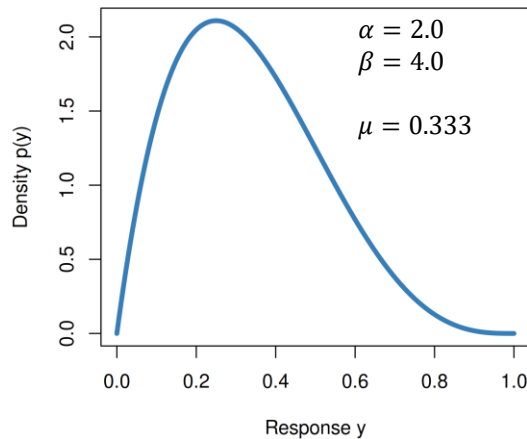
Inverse-logit ("logistic")

$$f = \text{logistic}(\eta) = \frac{\exp(\eta)}{1 + \exp(\eta)}$$

Beta distribution

$$y \sim \text{Beta}(\alpha, \beta)$$

Beta distribution



Continuous distribution, bounded between 0 and 1.

Complicated parameterization with 2 shape parameters α, β : Mean $\frac{\alpha}{\alpha+\beta}$. Sdev decreases with α and β

brms takes care of parameterization:

Mean μ and shape parameter ϕ generate α, β automatically

ATTN: Exact $y = 0$ or $y = 1$ not allowed.
→ Use zero- or one-inflated beta, or ordered beta.

Beta regression: model fitting

```
> brm(proportion ~ vegetation * predator,  
      family = Beta(link=logit),  
      prior = ... )
```

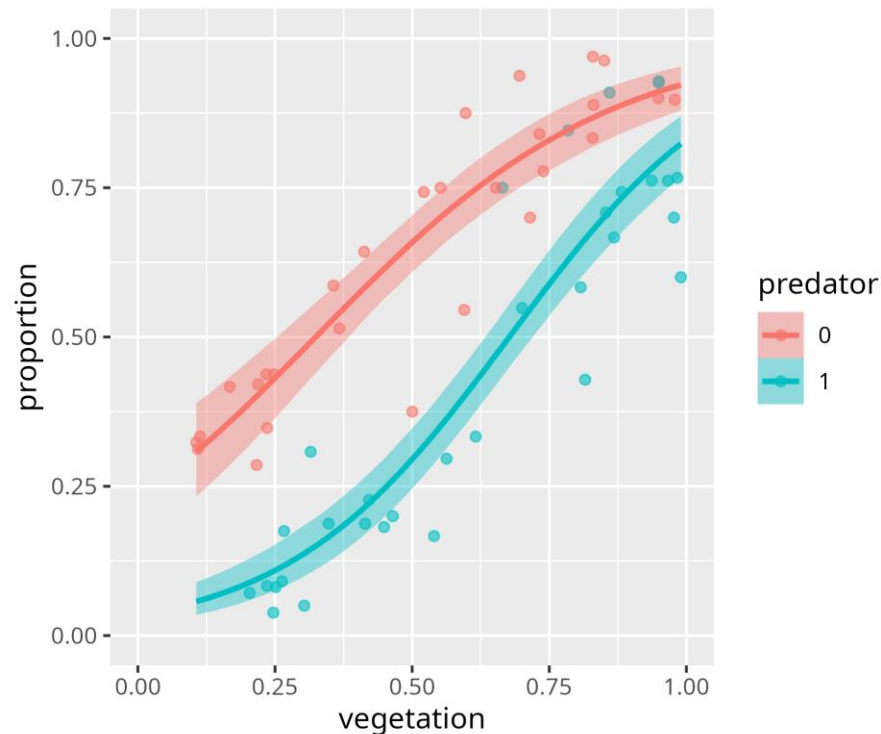
```
Family: beta  
Links: mu = logit; phi = identity  
Formula: proportion ~ scale(vegetation) * predator  
Data: df (Number of observations: 60)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.92	0.12	0.69	1.16	1.00	3467
scalevegetation	1.05	0.12	0.82	1.30	1.00	2425
predator1	-1.44	0.16	-1.76	-1.13	1.00	3824
scalevegetation:predator1	0.34	0.17	-0.01	0.67	1.00	2580

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	14.91	2.65	10.14	20.61	1.00	3450	2348



Results slightly different from Binomial regression

Additional scale parameter $\phi \sim 1/\text{sdev}$

Poisson regression

Poisson regression

Example: Abundance of trout in stream transects.

Recorded number of individuals per transect.

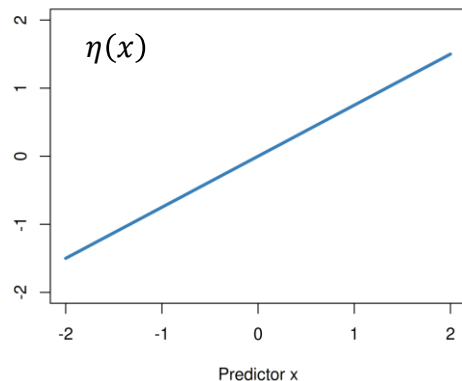
Measured temperature and concentration of a pollutant.

Q: Is pollutant concentration more harmful in warm streams?

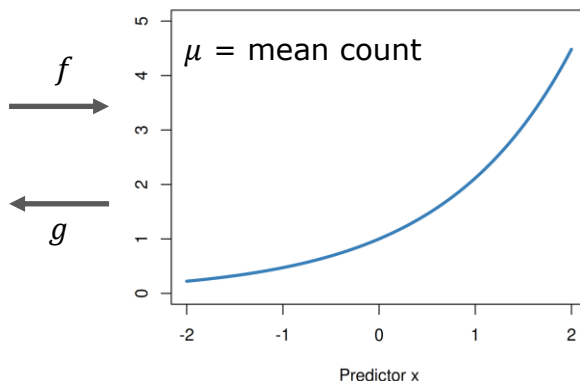


Binomial regression: overview

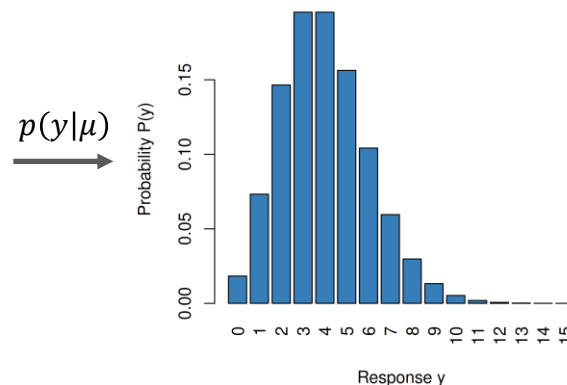
1: linear model



2: (inverse) link



3: residual distribution



Log link

$$g = \log(\mu)$$

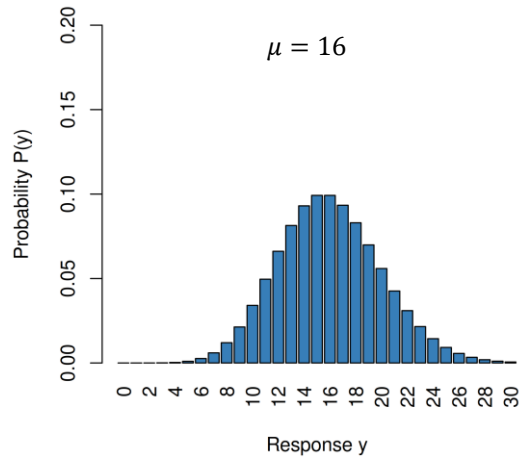
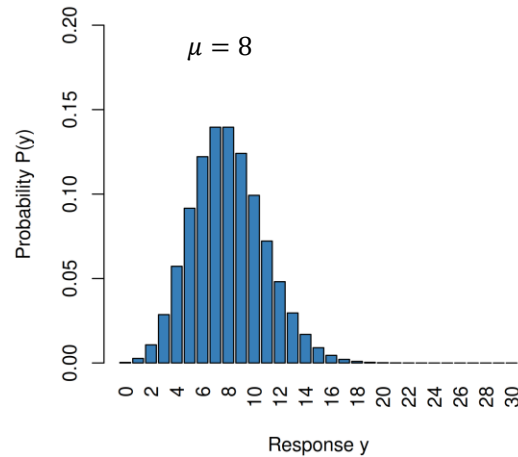
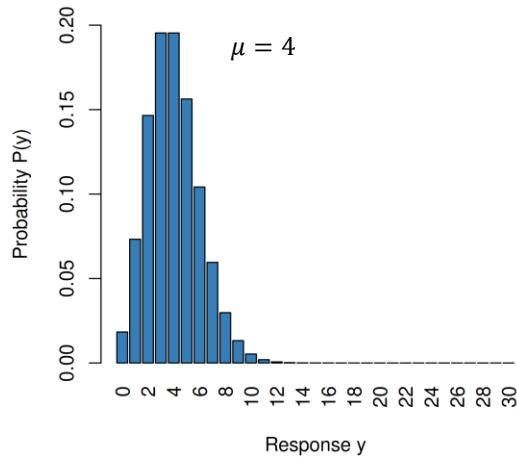
Inverse-link

$$f = \exp(\eta)$$

Poisson distribution

$$y \sim \text{Poisson}(\mu)$$

Poisson distribution



Discrete distribution, lower bound 0, no upper boundary → used for counting data

Mean: μ

Sdev: $\sqrt{\mu}$ (Variance equal to mean)

You make more errors counting many fish compared to when counting just a few fish

Poisson regression: interaction model

Deterministic part

(linear model & link)

$$\log(\mu) = b_0 + b_1 \cdot \text{temp} + b_2 \cdot \text{pollution} \\ b_3 \cdot \text{temp} \cdot \text{pollution}$$

Stochastic part

$$\text{abundance} \sim \text{Poisson}(\mu)$$

Priors

(scaled predictors!)

$$b_0 \sim \text{brms-default}$$

$$b_1 \sim \text{Normal}(0,1)$$

$$b_2 \sim \text{Normal}(-1,1) \text{ neg. pollution effect}$$

$$b_3 \sim \text{Normal}(0,1)$$

```
> brm(abundance ~ temperature * pollution,  
      family = poisson(link=log),  
      prior = ... )
```

	abundance	pollution	temperature
1	9	0.001	-0.418
2	12	-0.252	-1.650
3	17	-1.035	-1.266
4	2	2.749	-0.130
5	6	-0.377	1.982
6	7	0.318	2.096
7	12	-1.050	0.127
8	15	1.043	-1.989
9	16	-0.851	-0.295
10	14	-0.922	0.528
11	14	-0.503	-0.260
12	7	-0.740	0.437
13	6	1.391	-0.505
14	10	0.253	-0.980
15	14	0.263	-1.961

Poisson regression: additive model

```
> brm(abundance ~ temperature + pollution,  
      family = poisson(link=log),  
      prior = ... )
```

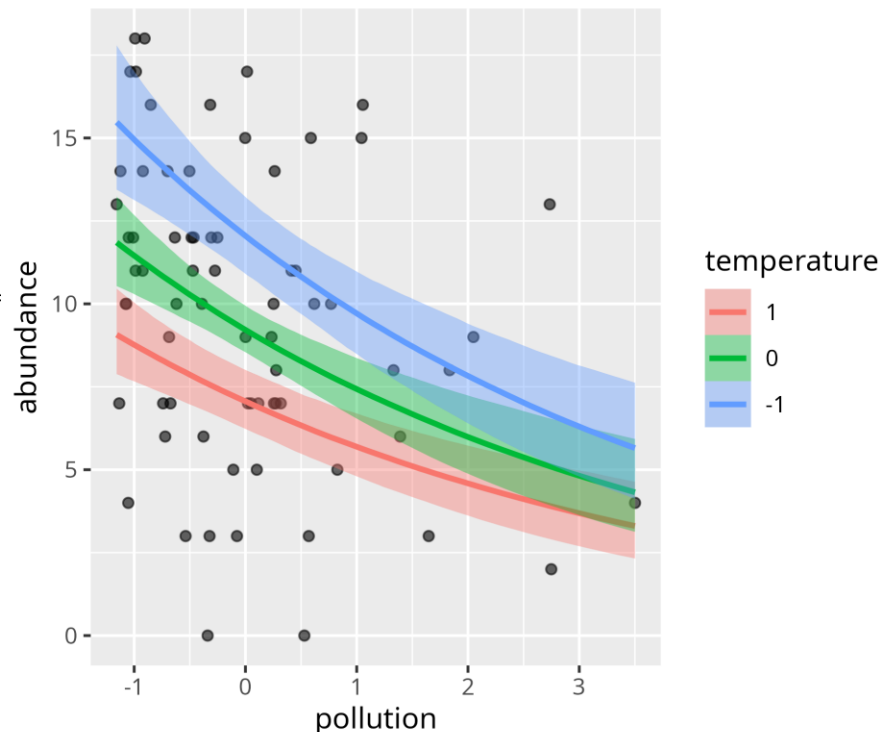
```
Family: poisson  
Links: mu = log  
Formula: abundance ~ temperature + pollution  
Data: df (Number of observations: 70)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin =  
       total post-warmup draws = 4000
```

Regression Coefficients:

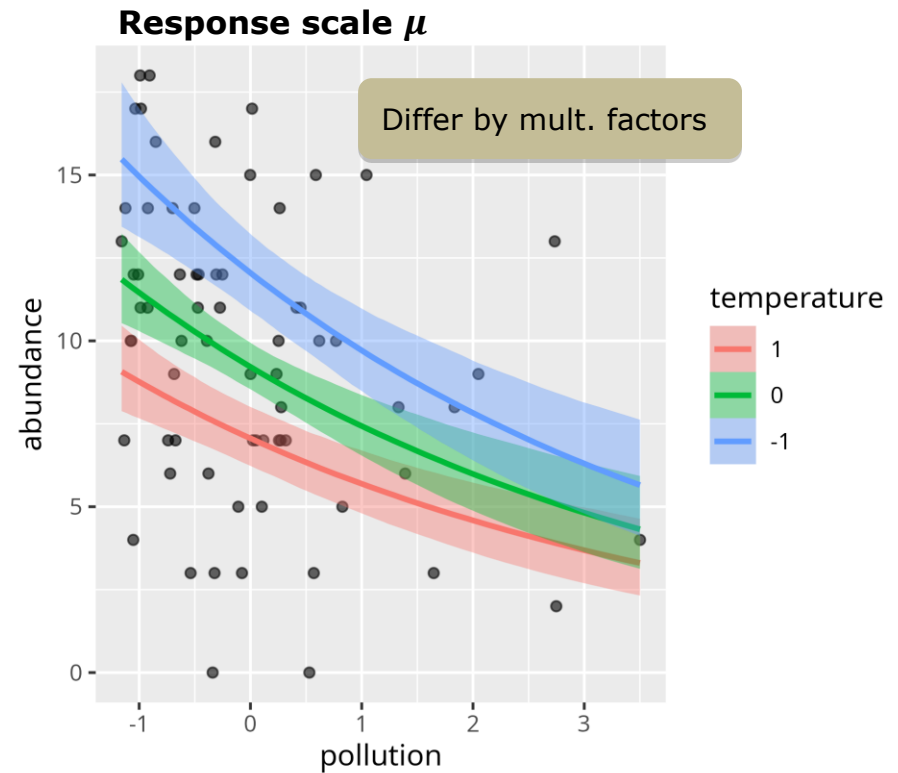
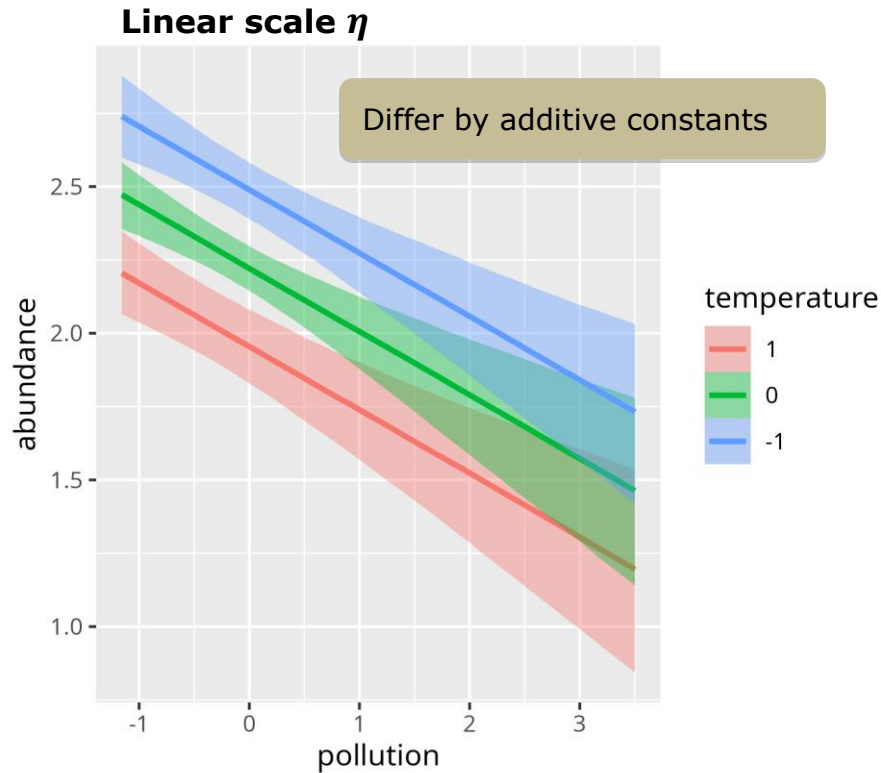
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	2.22	0.04	2.15	2.30	1.00	2807
temperature	-0.27	0.04	-0.35	-0.19	1.00	3102
pollution	-0.22	0.04	-0.30	-0.13	1.00	3512

→ Temperature and pollution both have negative effects on abundance

(slopes are on linear scale)



Poisson regression: additive model



Poisson regression: interaction model

```
> brm(abundance ~ temperature * pollution,  
      family = poisson(link=log),  
      prior = ... )
```

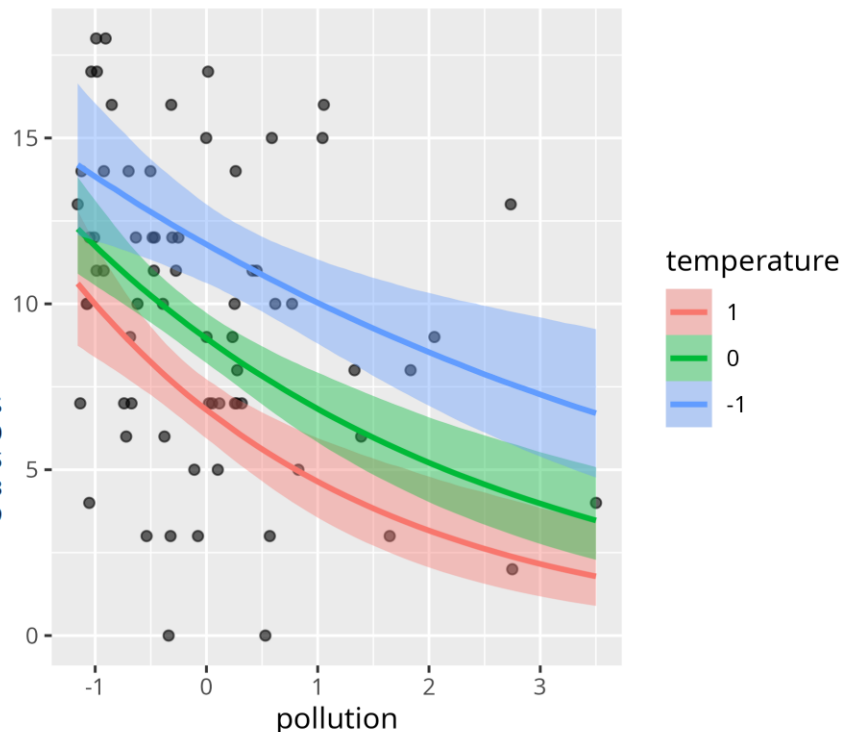
```
Family: poisson  
Links: mu = log  
Formula: abundance ~ temperature * pollution  
Data: df (Number of observations: 70)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
       total post-warmup draws = 4000
```

Regression Coefficients:

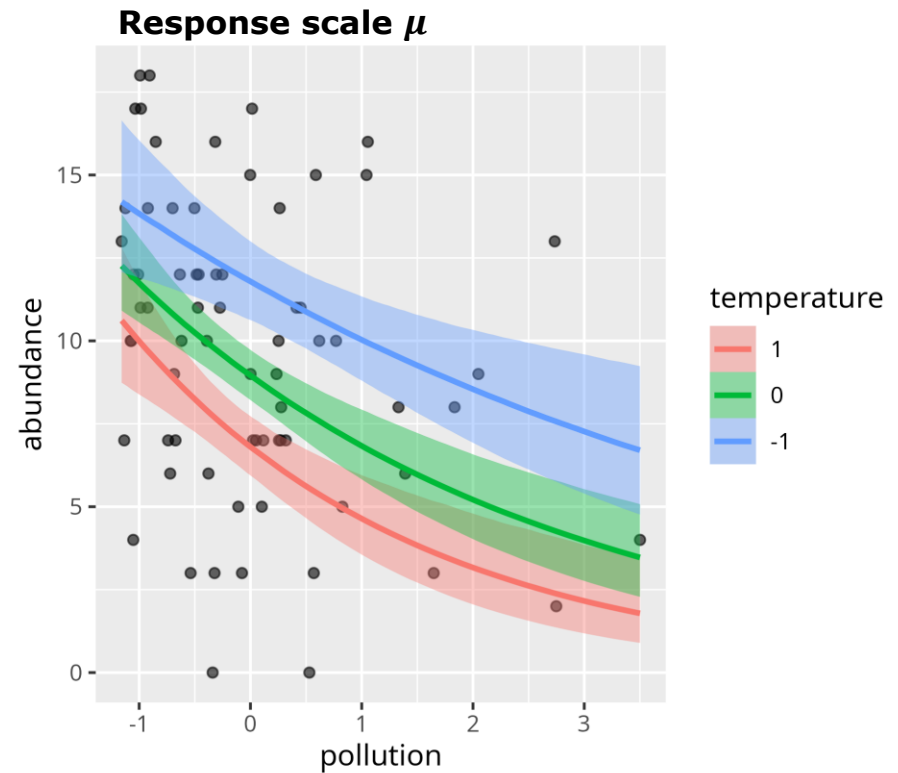
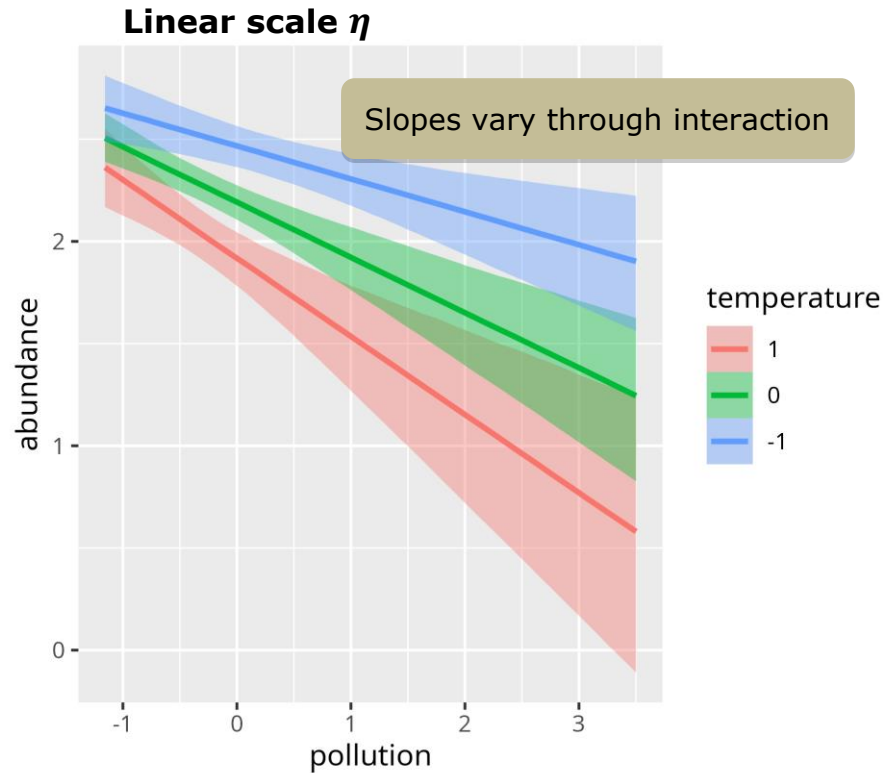
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	2.19	0.04	2.11	2.27	1.00	3490
temperature	-0.27	0.04	-0.36	-0.20	1.00	3545
pollution	-0.27	0.05	-0.38	-0.17	1.00	3378
temperature:pollution	-0.11	0.05	-0.21	-0.01	1.00	3419

Q: Is pollutant concentration more harmful in warm streams?

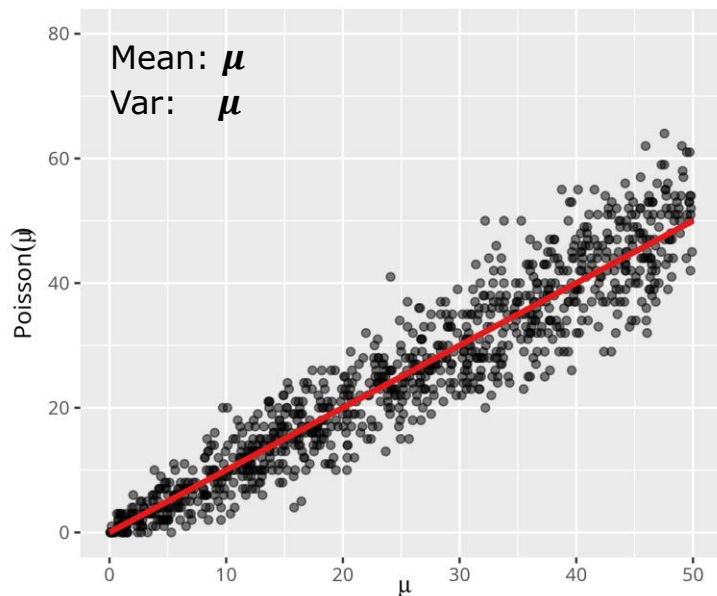
→ Yes, negative pollution effect gets stronger with temp.



Poisson regression: interaction model

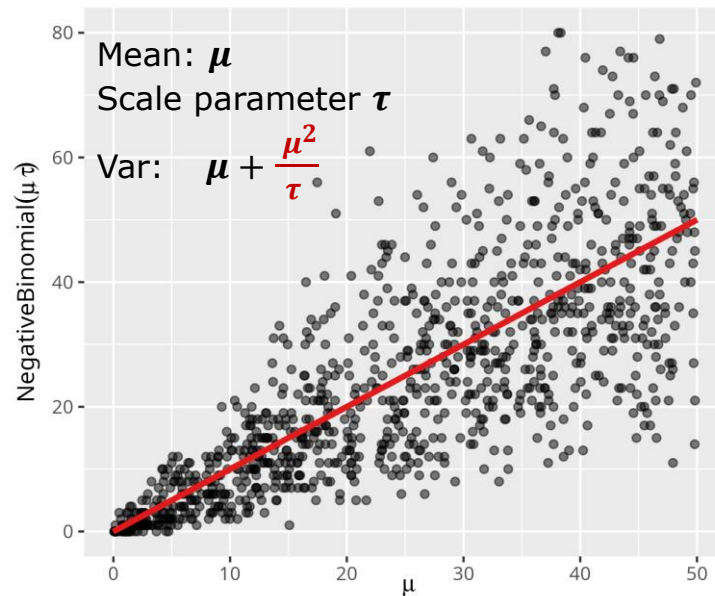


Poisson regression: Overdispersion



Often count data does not follow the mean-variance relation of Poisson

→ Use Negative Binomial distribution



τ small $\rightarrow \text{Var} \sim \mu^2$

τ big $\rightarrow \text{Var} \sim \mu$

> family=negbinomial(link=log)

*Continuous example:
Distributional model*

Distributional models

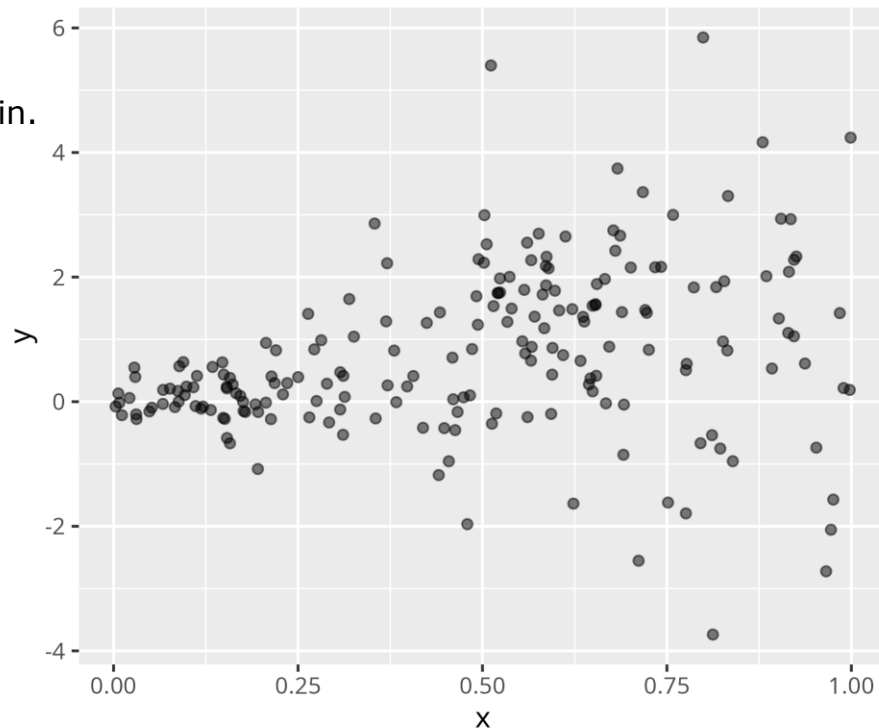
Data is clearly **heteroskedastic** (non-constant sdev)

Continuous response y → Can't use Poisson / Neg.Bin.

Linear regression line wanted → Can't use log-link

We can make **standard deviation dependent on x**

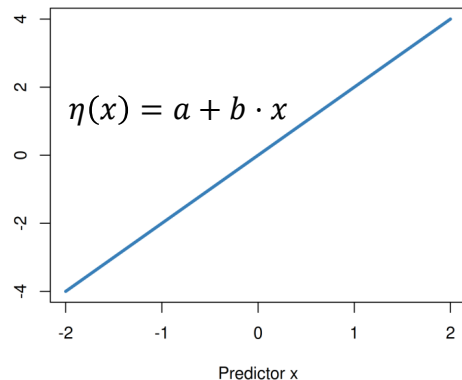
$$\rightarrow y \sim \text{Normal}(\mu(x), \sigma(x))$$



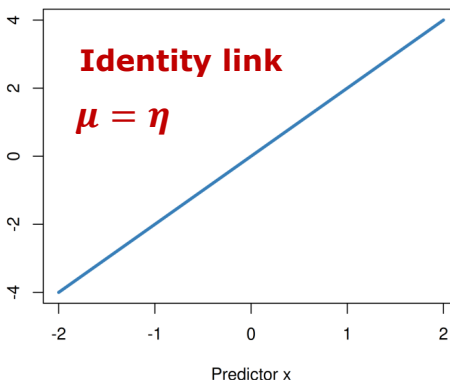
Distributional models

Mean model

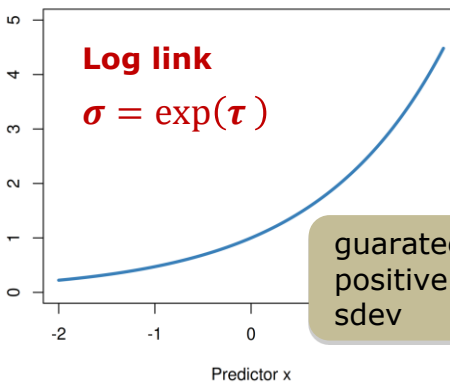
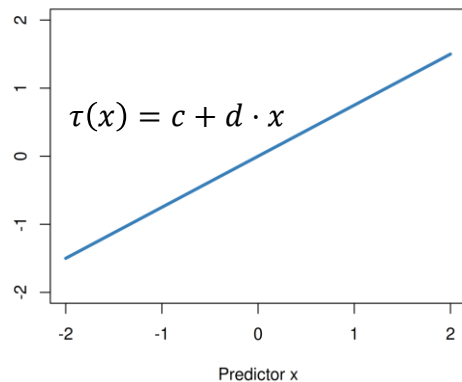
1: linear model



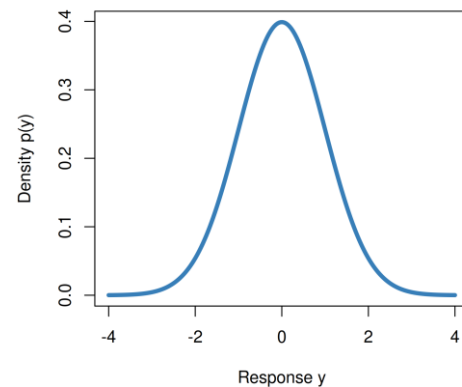
2: (inverse) link



Sdev model



3: residual distribution



$$y \sim \text{Normal}(\mu(x), \sigma(x))$$

Regular model fit

```
> fit_lm = brm( y ~ x )
```

```
Family: gaussian  
Links: mu = identity; sigma = identity  
Formula: y ~ x  
Data: df (Number of observations: 200)  
Draws: 4 chains, each with iter = 2000; warmu  
total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	0.09	0.18	-0.25	0.43
x	1.39	0.32	0.75	2.02

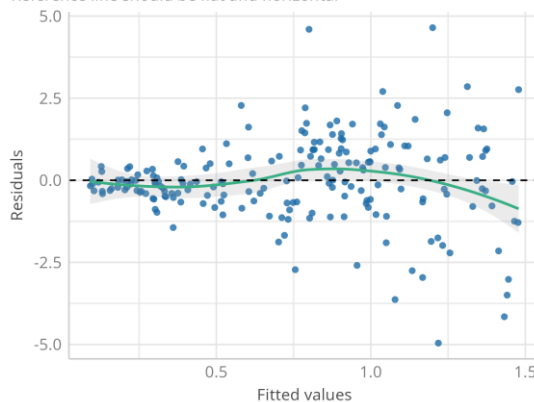
Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
sigma	1.29	0.07	1.17	1.42	1.00

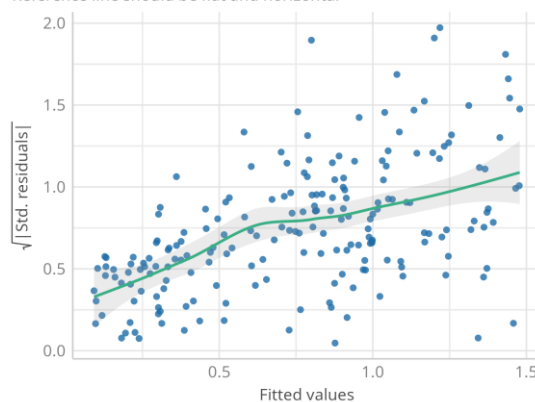
Bad model fit !

Some assumptions violated

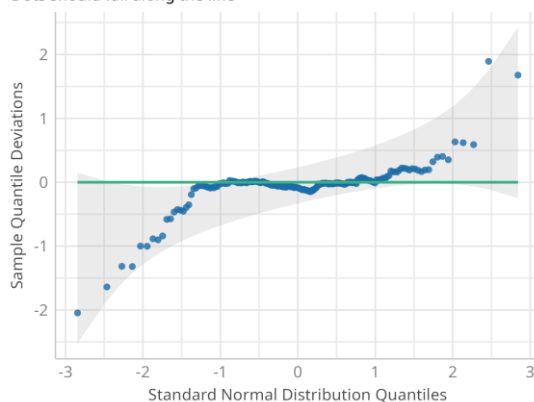
Linearity
Reference line should be flat and horizontal



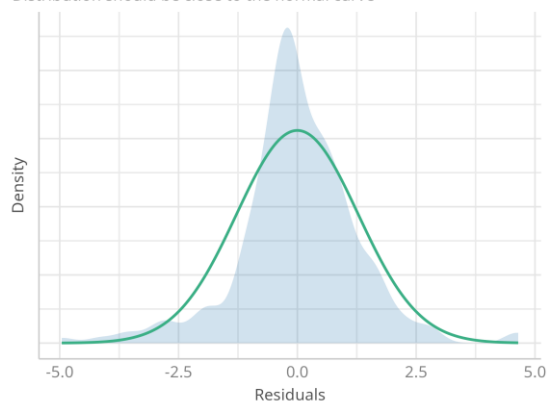
Homogeneity of Variance
Reference line should be flat and horizontal



Normality of Residuals
Dots should fall along the line

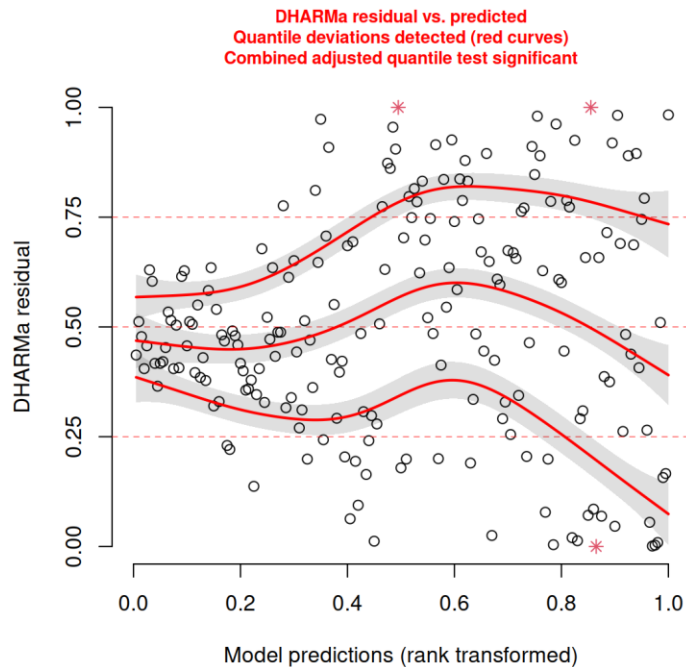
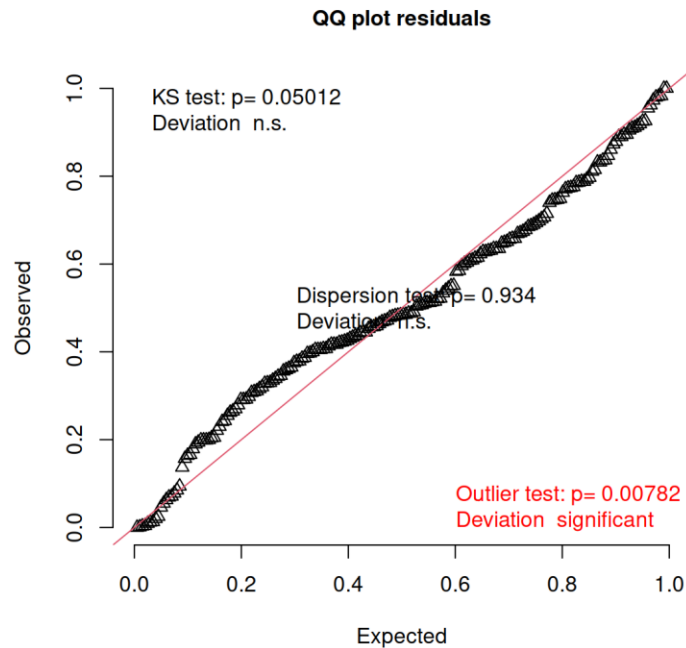


Normality of Residuals
Distribution should be close to the normal curve



Regular model fit

```
> DHARMA.helpers::dh_check_brms(fit_lm)
```



Distributional model fit

```
> fit_distr = brm( bf( y~x, sigma~x ),  
                  family = gaussian() )
```

Family: gaussian

Links: mu = identity; sigma = log

Formula: y ~ x

sigma ~ x

Data: df (Number of observations: 200)

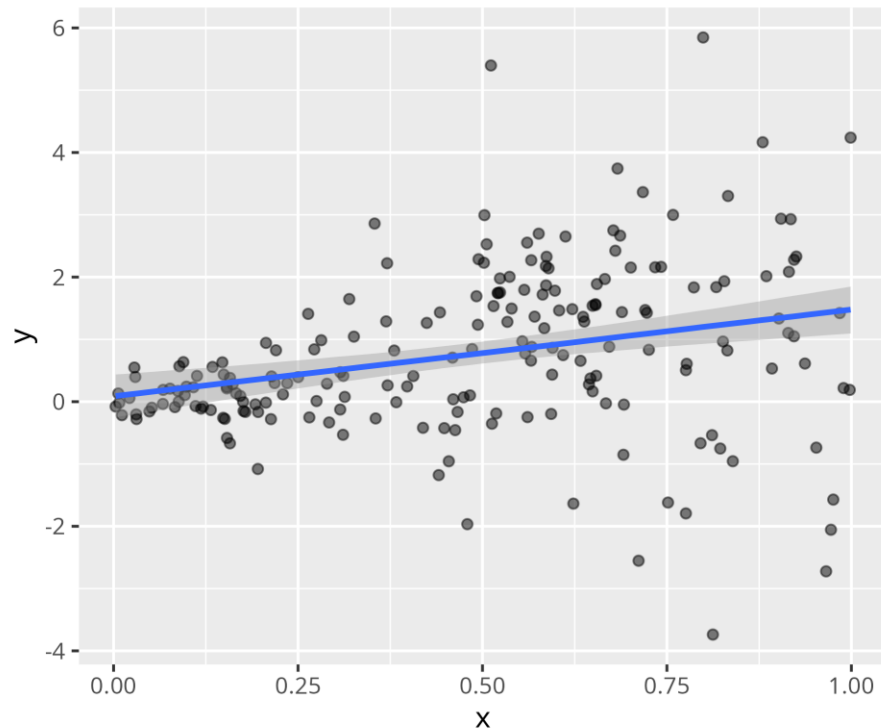
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Regression Coefficients:

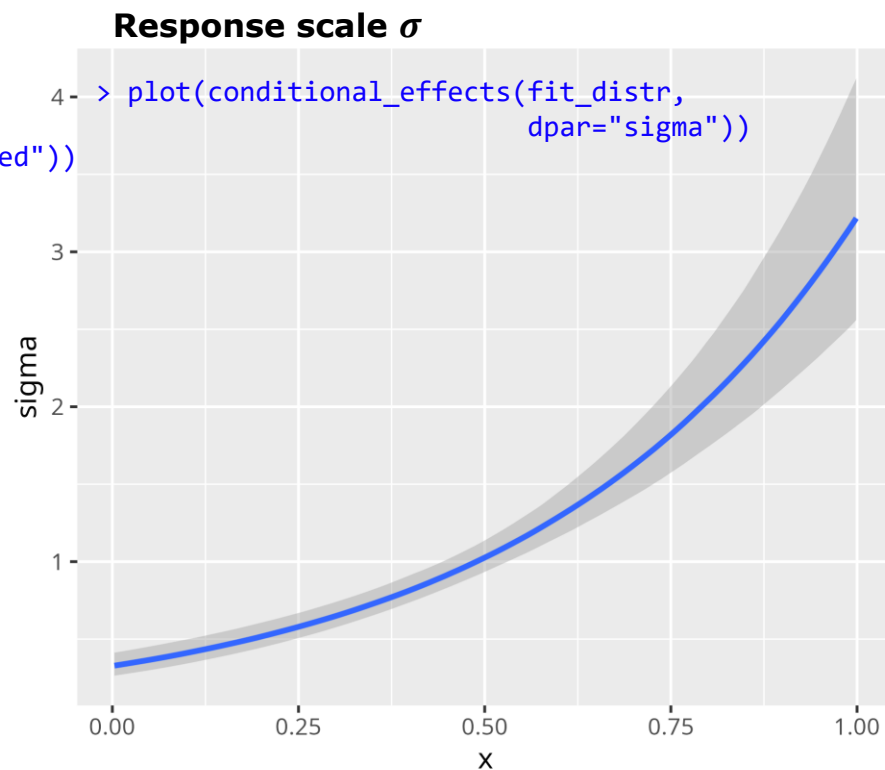
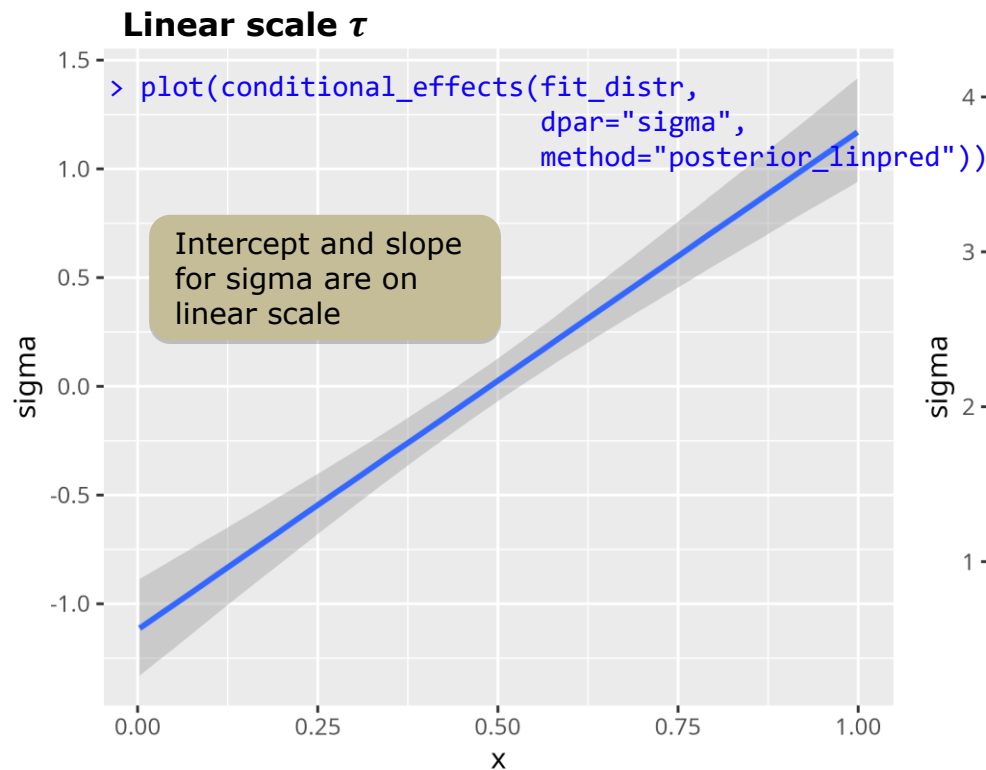
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	-0.09	0.07	-0.23	0.05	1.00	3776
sigma_Intercept	-1.12	0.11	-1.34	-0.89	1.00	3416
x	1.87	0.25	1.37	2.37	1.00	1987
sigma_x	2.29	0.21	1.88	2.72	1.00	3375

Log link for sigma as default

Slope in x substantially different from `fit_lm`

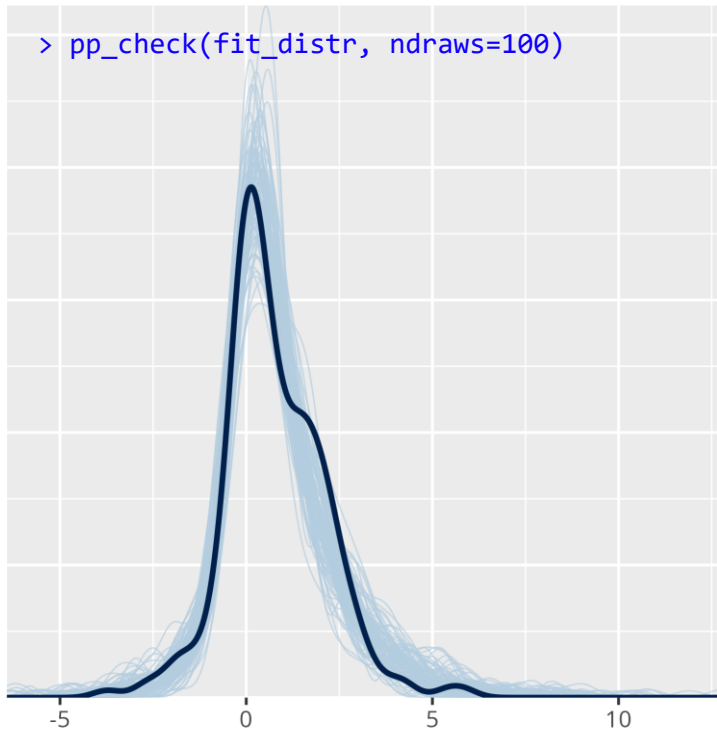


Distributional model fit

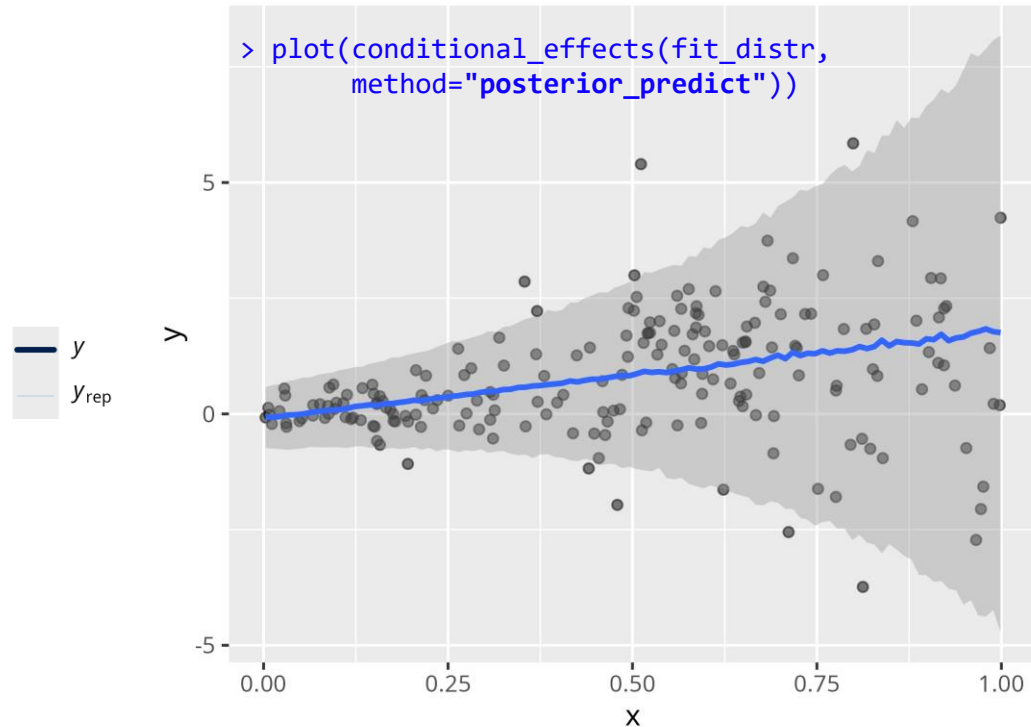


Distributional model: evaluation

```
> pp_check(fit_distr, ndraws=100)
```

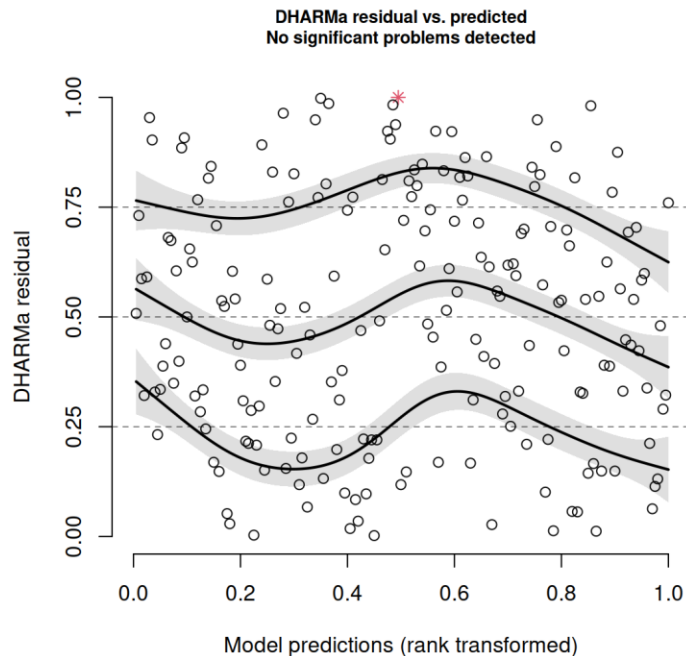
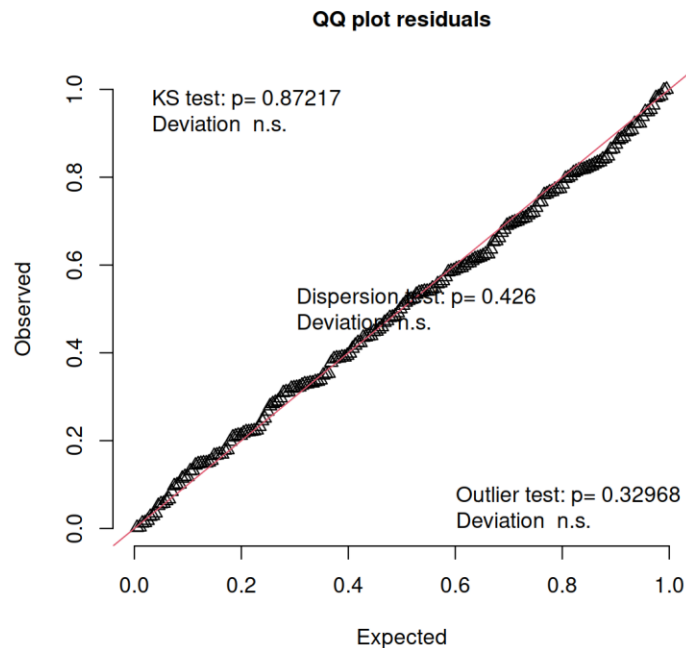


```
> plot(conditional_effects(fit_distr,  
  method="posterior_predict"))
```



Distributional model: evaluation

```
> DHARMA.helpers::dh_check_brms(fit_distr)
```



Distributional models: ANOVA

```
> fit.anova = brm( bf( y ~ 0+treatment,  
                      sigma ~ 0+treatment ),  
                  family = gaussian() )
```

Family: gaussian

Links: mu = identity; sigma = log

Formula: y ~ 0 + treatment

sigma ~ 0 + treatment

Data: df (Number of observations: 120)

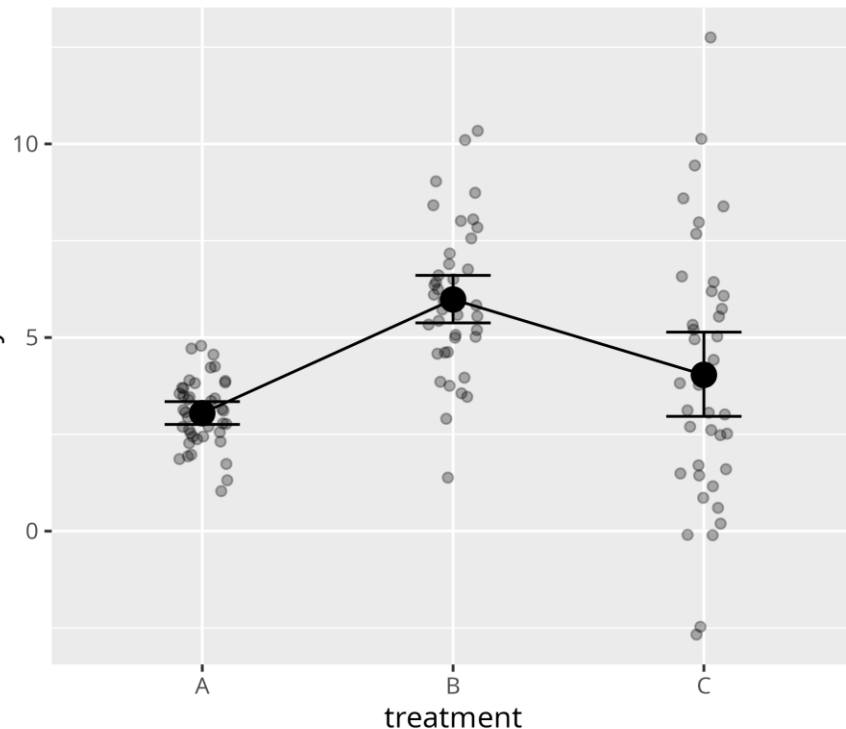
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1; >
total post-warmup draws = 4000

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
treatmentA	3.05	0.15	2.76	3.34	1.00	5711
treatmentB	5.99	0.31	5.38	6.61	1.01	5946
treatmentC	4.04	0.55	2.96	5.14	1.00	5347
sigma_treatmentA	-0.09	0.11	-0.31	0.14	1.00	5740
sigma_treatmentB	0.67	0.11	0.45	0.89	1.00	6186
sigma_treatmentC	1.23	0.11	1.02	1.46	1.00	6352

on linear scale.

→ $sdev = \exp(\sigma)$



*Beyond GLMs:
Nonlinear models*

Nonlinear model

Example: Microcosm experiment
Feeding rates of a small crustacean

“Functional response” $F(N) = \frac{aN}{1+ahN}$

Depends on prey density N

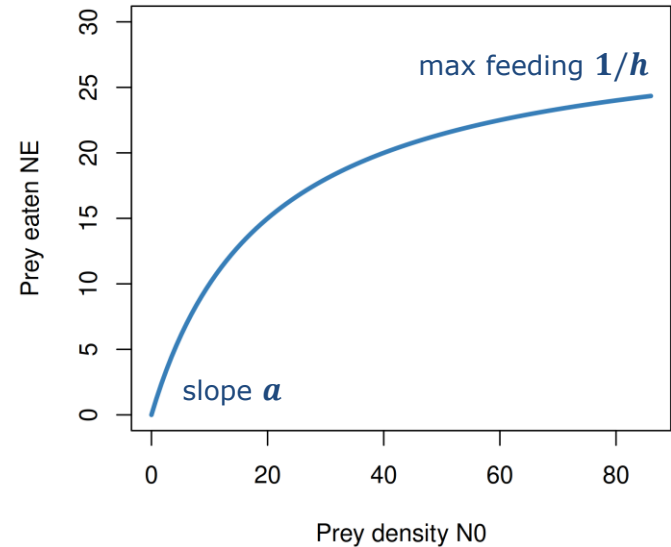
Want to estimate **attack rate** a and **handling time** h

Data:

Controlled feeding trials in the lab (1 hour)
at different prey densities N_0 .



Source: Mijel Decler CC BY-NC-SA 4.0
www.marinespecies.org/photogallery.php?album=717&pic=3616



Nonlinear model: fitting

```
FR.formula = bf( NE ~ a*N0/(1+a*h*N0),  
                 a ~ 1,  
                 h ~ 1,  
                 nl = TRUE)
```

→ the model formula
→ parameters do not depend
on other predictors
→ it's a nonlinear model

```
FR.priors = c(prior(exponential(1.0), nlpar="a", lb=0),  
              prior(exponential(1.0), nlpar="h", lb=0))
```

→ exp. distr. mean=1, sdev=1
and positive (lower bound 0)

```
fit.joint = brm(FR.formula,  
                 prior = FR.priors,  
                 family = poisson(link=identity),  
                 data = df)
```

→ response=counts, no log-link required
(FR model is always positive)

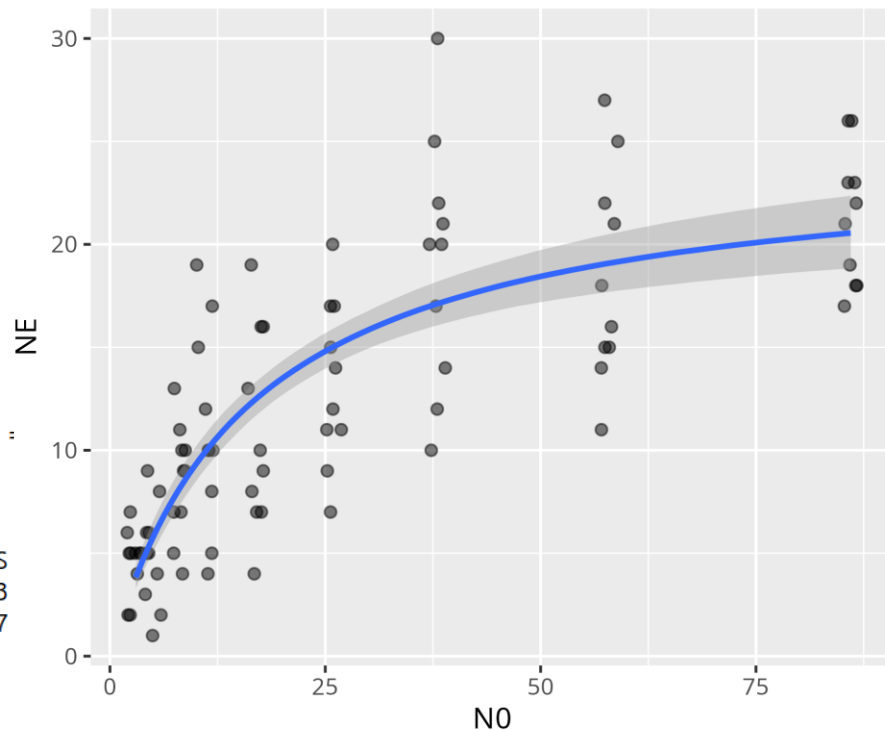
Nonlinear model: fitting

```
FR.formula = bf( NE ~ a*N0/(1+a*h*N0),  
                a ~ 1,  
                h ~ 1,  
                nl = TRUE)
```

```
Family: poisson  
Links: mu = identity  
Formula: NE ~ a * N0 / (1 + a * h * N0)  
         a ~ 1  
         h ~ 1  
Data: df (Number of observations: 90)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin =  
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
a_Intercept	1.52	0.15	1.26	1.85	1.01	1403
h_Intercept	0.04	0.00	0.04	0.05	1.00	1467



Nonlinear model: fitting

Include **categorical predictor** temperature (levels = low / high)

Q: Is feeding behavior different in warm experiments?

```
FR.formula = bf( NE ~ a*N0/(1+a*h*N0),  
                 a ~ 0 + temp,  
                 h ~ 0 + temp,  
                 nl = TRUE)
```

→ the model formula
→ **individual parameters at low / high temp.**
0 to avoid dummy-coding
→ it's a nonlinear model

```
FR.priors = c(prior(exponential(1.0), nlpar="a", lb=0),  
              prior(exponential(1.0), nlpar="h", lb=0))
```

→ exp. distr. mean=1, sdev=1
and positive (lower bound 0)

```
FR.fit.tmp = brm(FR.formula,  
                  prior = FR.priors,  
                  family = poisson(link=identity),  
                  data = df)
```

→ response=counts, no log-link required
(FR model is always positive)

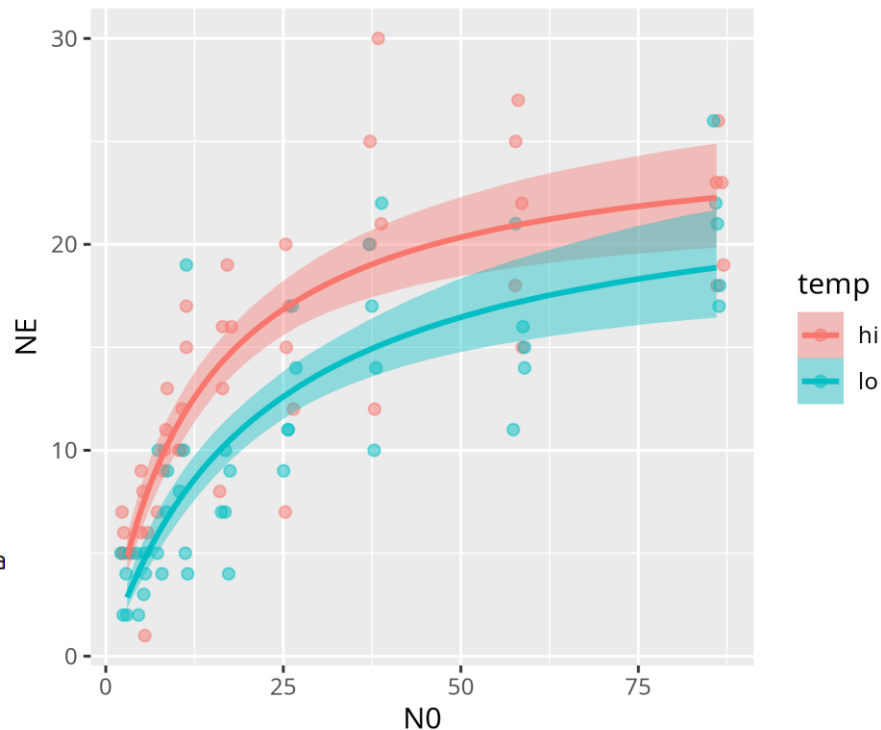
Nonlinear model: fitting

```
FR.formula = bf( NE ~ a*N0/(1+a*h*N0),  
                a ~ 0 + temp,  
                h ~ 0 + temp,  
                nl = TRUE)
```

```
Family: poisson  
Links: mu = identity  
Formula: NE ~ a * N0/(1 + a * h * N0)  
        a ~ 0 + temp  
        h ~ 0 + temp  
Data: df (Number of observations: 90)  
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin  
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Ta
a_temphi	2.00	0.26	1.56	2.57	1.00	2515	
a_templo	1.10	0.15	0.84	1.44	1.00	2277	
h_temphi	0.04	0.00	0.03	0.05	1.00	2663	
h_templo	0.04	0.00	0.03	0.05	1.00	2381	



Nonlinear model: fitting

```
> hypothesis(FR.fit.tmp,  
             c("a_temphi > a_templo",  
               "h_temphi > h_templo" ))
```

Hypothesis Tests for class b:

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
1	(a_temphi)-(a_tem... > 0	0.9	0.30	0.43	1.42	1332.33	1.00	*
2	(h_temphi)-(h_tem... > 0	0.0	0.01	-0.01	0.01	0.41	0.29	

'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.

'*': For one-sided hypotheses, the posterior probability exceeds 95%;

for two-sided hypotheses, the value tested against lies outside the 95%-CI.

Posterior probabilities of point hypotheses assume equal prior probabilities.

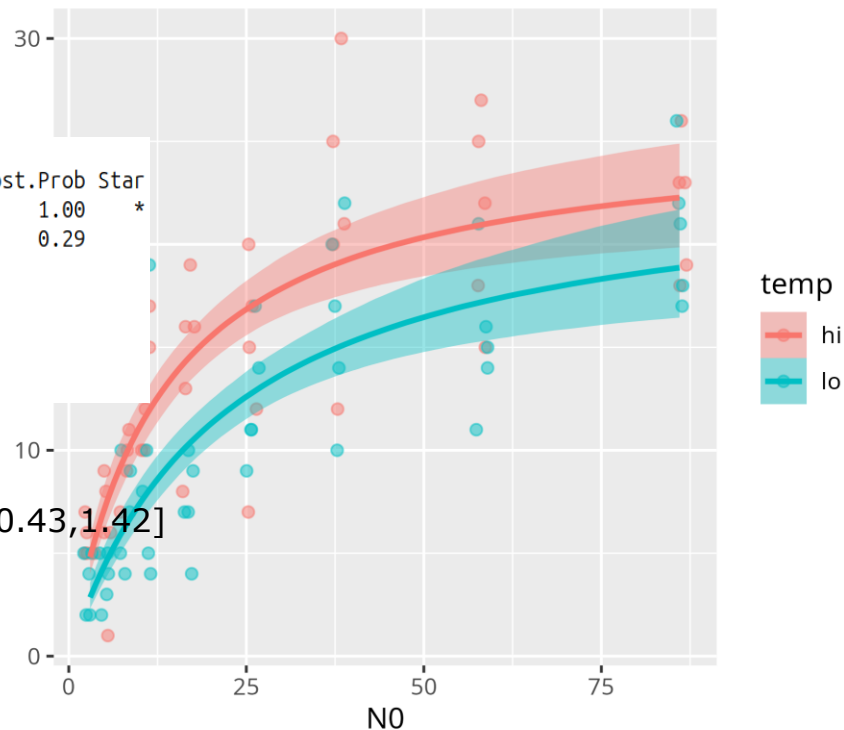
Q: Is feeding behavior different in warm experiments?

→ Yes: „warm“ attack rates higher, mean diff. = 0.9 [0.43,1.42]
But no difference in handling time (max feeding)

```
> LOO(FR.fit.tmp, FR.fit)
```

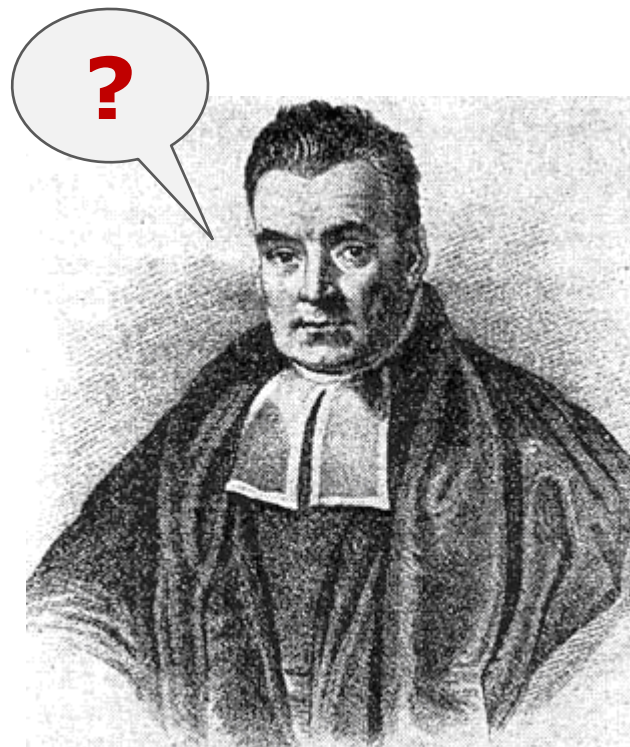
Model comparisons:

	elpd_diff	se_diff
FR.fit.tmp	0.0	0.0
FR.fit	-11.0	5.6



What's Bayesian about it?

- Exact quantification of parameter uncertainty.
Even more important when moving beyond simple LMs.
- Don't use parameters' point estimates for prediction
→ Use full posterior predictive distribution
→ emmeans for contrasts
- Works well even for very small datasets
- Not limited to GLM framework
- Put models into models
`bf( y ~ x1+x2, sigma ~ x1 )` etc ...
- No extra packages needed, do it all in brms

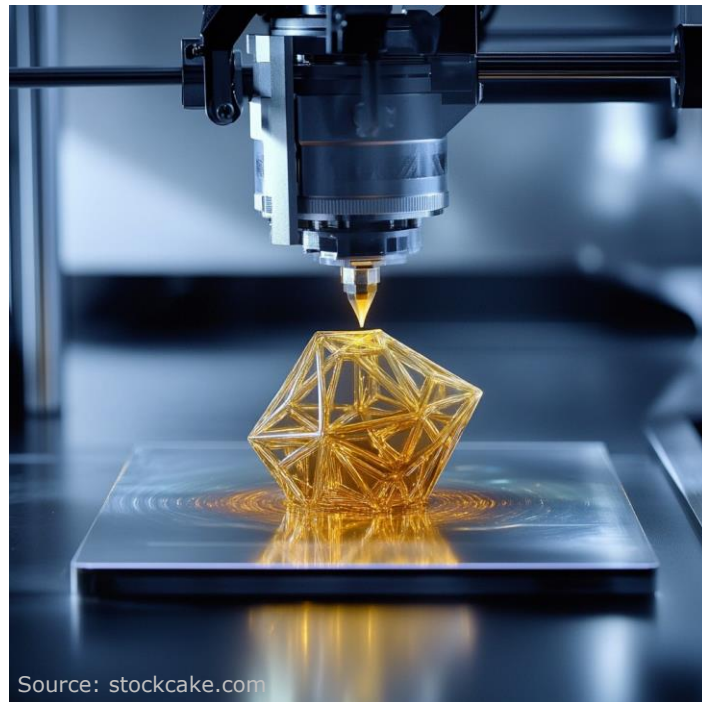


Source: Wikipedia

Summary

- brms highly flexible, can fit vast amount of GLMs. Even nonlinear models!
- Bayesian stats not limited to GLM, but linear part in GLM is useful and interpretable
- Data should indicate which GLM (distribution & link) to use
- Be careful with priors when using link functions. Scaled / mean-centered predictors are your friend
- Use appropriate plots for model evaluation (DHARMa)
- ATTN: Don't do model comparison (loo) for discrete vs. continuous residual distribution
E.g. Poisson vs. Gaussian

The Bayesian 3D printer



Source: stockcake.com

Further reading

Buckley, Y. M. (2015). Generalized linear models. In Fox, G. A. , Negrete-Yankelevich, S. and Sosa, V. J. (eds). *Ecological Statistics: Contemporary theory and application*. *Oxford Academic*.
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Gelman, A., Hill, J., & Vehtari, A. (2020). *Regression and Other Stories*. *Cambridge University Press*.
<https://doi.org/10.1017/9781139161879> [Part 3]

Inchausti, P. (2023). *Statistical Modeling With R: a dual frequentist and Bayesian approach for life scientists*. *Oxford University Press*. [Chapters 8-12]

Kery, M. & Kellner, F. (2024): *Applied Statistical Modelling for Ecologists*. *Elsevier*. [Chapters 11-16]

Warton, D. (2022). *Eco-Stats: Data Analysis in Ecology*. *Springer (Methods in Statistical Ecology)* [Chapter 10]

Warton, D., Lyons, M., Stoklosa, J., & Ives, A. R. (2016). Three points to consider when choosing a LM or GLM test for count data. *Methods in Ecology and Evolution*, 7(8), 882–890. <https://doi.org/10.1111/2041-210X.12552>